

SPARSE SYZGY COLUMNS AND LECH'S CONJECTURE IN EQUAL CHARACTERISTIC

ANONYMOUS

ABSTRACT. Let $(R, \mathfrak{m}) \rightarrow (S, \mathfrak{n})$ be a flat local map of Noetherian local rings containing fields of the same characteristic. We prove Lech's inequality $e(R) \leq e(S)$ by establishing a length estimate for finite-length modules over equicharacteristic regular local rings. The estimate bounds each Betti number in terms of the depth of the columns in a minimal free resolution, with an error depending only on the number of exceptional columns. Its constant is independent of all ranks and grading shifts.

The graded argument combines the Eisenbud–Erman categorical pairing, Eisenbud–Schreyer supernatural bundles, and a positive interpolation identity. Prescribed-jet globalization and periodic degree boxes transfer the estimate to power-series rings. A coordinate-power argument then controls the errors introduced by scalar-nullhomotopic comparison cones and asymptotically negligible positive homology.

For a complete local ring D of characteristic $p > 0$ and dimension d , a free parameter lattice of rank $e(D)$ provides the required comparison for Frobenius pullbacks. Together with the Frobenius homology estimate of Roberts and its extension by Hochster–Huneke, this yields

$$\binom{d}{i} \chi_{\infty}^D(F) \geq e(D) \operatorname{rank}_D F_i \quad (0 \leq i \leq d)$$

for every minimal short free complex F over D . Passage to completion gives the same inequality without the completeness hypothesis. The reduction of Iyengar–Ma–Walker gives Lech's inequality in positive characteristic; the specialization argument recorded by Ma, who credits its strategy to Hochster, gives equal characteristic zero.

1. INTRODUCTION

For a Noetherian local ring $(R, \mathfrak{m})$, let $e(R) = e_{\mathfrak{m}}(R)$ denote its Hilbert–Samuel multiplicity. Lech's conjecture asserts that a flat local map $(R, \mathfrak{m}) \rightarrow (S, \mathfrak{n})$ satisfies

$$e(R) \leq e(S). \tag{1.1}$$

1.1. Main results.

Theorem 1.1. *Let $(R, \mathfrak{m}) \rightarrow (S, \mathfrak{n})$ be a flat local map of Noetherian local rings. Assume that R and S contain fields of the same characteristic. Then*

$$e(R) \leq e(S).$$

The positive-characteristic argument proves a stronger homological inequality. A *short free complex* over a d -dimensional local ring is a complex of finite free modules in degrees $0, \dots, d$ with finite-length homology, not all zero. For such a complex in characteristic $p > 0$, write $\chi_{\infty}^D(F)$ for its Dutta multiplicity; the normalization is given in (8.1).

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Theorem 1.2. *Let $(D, \mathfrak{m}, k)$ be a complete Noetherian local ring of characteristic $p > 0$ and dimension d . Let*

$$F : 0 \longrightarrow F_d \longrightarrow \cdots \longrightarrow F_0 \longrightarrow 0$$

be a minimal complex of finite free D -modules whose homology has finite length. Assume $F_0 \neq 0$. Then, for every $0 \leq i \leq d$,

$$\binom{d}{i} \chi_\infty^D(F) \geq e(D) \operatorname{rank}_D F_i. \quad (1.2)$$

No Cohen–Macaulay, reducedness, or equidimensionality hypothesis is imposed on D . Completeness can also be removed, as recorded in Corollary 8.2. Minimality is necessary when the right side is written using term ranks; equivalently, one can use the Betti numbers of a short complex. The case $i = 0$ and $F_0 = D$ gives $\chi_\infty^D(F) \geq e(D)$. The proof of Iyengar–Ma–Walker [16, Proposition 8.3] associates to a suitably reduced flat local map a ring T and a minimal short complex X with

$$e(T) = e(R), \quad X_0 = T, \quad \chi_\infty^T(X) = e(S).$$

Thus the row-zero case of Theorem 1.2 is exactly the input needed for their reduction.

The main estimate concerns columns in a resolution over a regular local ring. After choosing bases, call a source column *N-deep* if all its entries belong to $\mathfrak{m}^N$. Columns that do not satisfy this condition are *exceptional*.

Theorem 1.3 (sparse-column inequality). *Let $(A, \mathfrak{m}, k)$ be an equicharacteristic regular local ring of dimension $d \geq 1$, let M be a finite-length A -module, and let $N \geq 1$ be an integer. Choose bases of the minimal free resolution of M . Suppose that at most B source columns, counted over all positive differentials, are not N -deep. Then, for every $0 \leq r \leq d$,*

$$\ell_A(M) \geq N^d \frac{\beta_r^A(M)}{\binom{d}{r}} - C_d N^d B, \quad C_d = \frac{d^{d-1}}{(d-1)!}. \quad (1.3)$$

Consequently,

$$\ell_A(M) \geq \left(\frac{N}{2}\right)^d \sum_{r=0}^d \beta_r^A(M) - C_d N^d B. \quad (1.4)$$

The constant depends only on d .

The main term is attained by $A/(x_1^N, \dots, x_d^N)$ when $x_1, \dots, x_d$ is a regular system of parameters. The feature needed here is the error term: it counts exceptional columns without requiring any bound on their entries, on the ranks of the resolution, or on shifts in a grading. This uniformity permits the estimate to be applied to the moving resolutions used in the Frobenius argument.

1.2. Relation to earlier work. Lech proved this when $\dim R \leq 2$ and when the closed fiber $S/\mathfrak{m}S$ is a complete intersection, as well as the general estimate $e(R) \leq (\dim R)!e(S)$ [19, 20]. The construction of Herzog, Ulrich, and Backelin implies the conjecture when R is a strict complete intersection [15]. In equal characteristic, Ma proved the dimension-three case and the bound $e(R) \leq \max\{1, d!/2^d\}e(S)$, where $d = \dim R$ [21]. He later proved the conjecture when R is a standard graded algebra over a perfect field localized at its homogeneous maximal ideal [22]. Meng proved a complementary case for maps of equal dimension whose completed target and extended maximal ideal are obtained by completing a standard graded algebra and a homogeneous ideal, respectively [23, Theorem 5.13].

The Length Conjecture of Iyengar–Ma–Walker predicts that a nonzero finite-length module of finite projective dimension over a local ring D has length at least $e(D)$ [16, Conjecture 1]. Their related short-complex conjecture asks for $\chi_\infty(F) \geq e(D)$ [16, Conjecture 8.1]. They establish stronger binomial bounds for finite-length modules of finite projective dimension over strict

complete intersections [16, Theorem 3.8] and for short complexes over standard graded rings of positive characteristic localized at their homogeneous maximal ideals [16, Corollary 7.2]. Their later work proves a Betti-cone comparison for graded rings admitting linear Noether normalizations over infinite perfect fields [17, Theorem 6.2]. KC and Soto Levins prove the length bound for modules admitting a Serre lift to an unramified regular local ring [18]. These results concern related lower bounds; the conclusion proved here is the Dutta-multiplicity bound (1.2), extended to arbitrary local rings of positive characteristic in Corollary 8.2. For a finite free resolution F of a finite-length module M , one has $\chi(F) = \ell_D(M)$; the present argument does not supply the additional comparison with $\chi_\infty(F)$ needed to infer the ordinary Length Conjecture.

The graded part of our proof uses the Boij–Söderberg framework introduced by Boij and Söderberg [4] and developed by Eisenbud and Schreyer [8, Theorems 0.1 and 0.2]. More specifically, we use the supernatural bundles of Eisenbud–Schreyer [8, Theorems 6.1 and 6.4] and the categorical pairing of Eisenbud–Erman, including its Betti-table formula and the finite-length homology of the paired complex [7, Definition 2.1, Theorem 2.3, and Proposition 2.6]. The one-variable truncated Euler inequalities and the choice of roots by omitting adjacent indexed nodes have close antecedents in Eisenbud–Erman [7, Definition 4.1 and Proposition 4.2] and Eisenbud–Schreyer [8, proof of Theorem 7.1], respectively. The columnwise refinement of that pairing, the resulting quantitative error bound, and the positive interpolation identity are proved in Section 3. The moment identities used there are the Peskine–Sziro multiplicity equations [24]; the pure-resolution specialization is due to Herzog–Kühl [14]. The filtration theorem of Eisenbud–Erman–Schreyer concerns pure degree sequences with separation hypotheses [9, 10]; we do not use that filtration theorem in the column-counting argument.

For the local transfer, the prescribed-generator argument adapts the method behind the Forster–Swan theorem [11, 30], and Quillen–Suslin supplies freeness of the final polynomial syzygy [25, 29]. The strict parameter lattice is constructed from classical graded Noether normalization and multiplicity theory. Its role here is to turn those classical constructions into a comparison with controlled differential depth and one fixed scalar nullhomotopy. The Frobenius homology estimate needed here is the case due to Roberts [26] of Hochster–Huneke’s more general theorem [13, Theorem 6.2]; we use the formulation and priority statement in Bhatt–Hochster–Ma [3, Theorem 5.2 and the preceding paragraph]. The final flat-map and characteristic-zero reductions are due to the cited work of Ma and Iyengar–Ma–Walker [21, 16]. The argument does not require the existence of weakly lim Ulrich sequences; Yhee shows that such sequences need not exist for complete local domains [32].

1.3. Structure of the proof. There are four stages, with two distinct limiting procedures.

The graded estimate. For each adjacent pair of homological rows, a translated supernatural bundle selects a window of N grading shifts. A deep column contributes nothing to the relevant map in that window, while each exceptional column appears for at most N translations. Summing these tests and using a positive interpolation identity gives the graded form of (1.3) in Section 3.

Transfer to local resolutions. Section 4 constructs an exact polynomial resolution of a finite direct sum of a given local module while preserving the selected column depths. Section 5 compares periodic and open degree boxes. Periodic boxes preserve exactness; open boxes retain a grading. A correction of total free rank $O(L^{d-1})$ turns an open box into a resolution of a finite-length graded module. Apply the graded estimate before taking $L \rightarrow \infty$: a small-rank correction can carry a leading degree moment, so its shifts cannot be discarded. This limit is taken for each fixed module and proves Theorem 1.3 with the uniform constant C_d .

From deep complexes to finite-length modules. Section 6 starts with a complex of fixed row ranks b_i and differential depth $N_q \sim \rho q$. Its comparison with a finite-length module is controlled by a scalar-nullhomotopic cone and small positive homology. Restriction along $x_j \mapsto x_j^{t_q}$, for a

suitable $t_q = o(q)$, makes these errors negligible relative to t_q^d . After minimization the resulting resolution has

$$\beta_i = t_q^d b_i + o(t_q^d), \quad \text{depth } n_q \sim \rho q / t_q, \quad B_{n_q} = o(t_q^d).$$

Thus $(n_q t_q)^d \sim \rho^d q^d$, and the local sparse-column inequality gives the required normalized length bound. This is the second limit; the degree-box construction has already been completed.

Frobenius and the final reductions. The lattice in Section 7 has rank $e(D)$ and gives $b_i = e(D) \text{rank}_D F_i$, depth $q - O(1)$, and one fixed scalar controlling the comparison cone. The Roberts estimate, in the stated Hochster–Huneke formulation, makes positive Frobenius homology negligible at scale q^d . Section 8 then proves Theorem 1.2, and Section 9 applies the literature reductions to obtain Theorem 1.1. Section 10 identifies the uses of equal characteristic. The argument supplies no corresponding conclusion in mixed characteristic.

2. CONVENTIONS AND ELEMENTARY HOMOLOGICAL ESTIMATES

We first record the homological estimates used to measure errors in a free complex. There are two kinds of error in the later argument: finite-length homology, measured by Lemma 2.2, and a comparison cone on which a scalar is nullhomotopic, measured by Lemma 2.5. Lemma 2.4 explains how these estimates control the columns of an actual minimal resolution.

2.1. Complexes, Betti numbers, and column orders. Complexes are homological: their differentials lower degree. We use the shift convention $X[1]_i = X_{i-1}$, with differential $-d_X$. For a chain map $f : X \rightarrow Y$, our mapping cone is

$$\text{Cone}(f)_i = Y_i \oplus X_{i-1}, \quad d_{\text{Cone}(f)}(y, x) = (d_Y y + f(x), -d_X x).$$

It fits into the degreewise split exact sequence $0 \rightarrow Y \rightarrow \text{Cone}(f) \rightarrow X[1] \rightarrow 0$. A *short complex* over a d -dimensional local ring is a complex of finite free modules supported in degrees $0, \dots, d$ whose homology modules have finite length and are not all zero.

We use derived tensor products when a complex is not already free. Concretely, $k \otimes_A^{\mathbf{L}} C$ is computed by replacing C by a free resolution of the complex before tensoring with k . A *perfect complex* is one admitting a bounded resolution by finite free modules; over a local ring we call such a resolution a *finite free model*. For a bounded complex C with finitely generated homology over $(A, \mathfrak{m}, k)$, set

$$\beta_i^A(C) = \dim_k H_i(k \otimes_A^{\mathbf{L}} C), \quad \beta_A^{\text{tot}}(C) = \sum_i \beta_i^A(C).$$

When C is a bounded free complex, ordinary tensor with k computes these groups. When C is a module in degree zero, they are its usual Tor groups. The total is finite for a perfect complex. On a minimal free model all differentials vanish modulo $\mathfrak{m}$, so $\beta_i^A(C)$ is the free rank in degree i . We omit A when the ring is clear.

Over a standard graded polynomial ring S with homogeneous maximal ideal $\mathfrak{m}$, we use the same definition with $k = S/\mathfrak{m}$ and graded free models. The number $\beta_i^S(C)$ is the total rank in homological degree i , summed over internal shifts. Localizing a minimal graded model at $\mathfrak{m}$ gives a minimal model over $S_{\mathfrak{m}}$. Thus these Betti numbers agree with the local ones. For modules supported at $\mathfrak{m}$, localization also preserves length; in particular, the local length estimates below apply to finite-length graded modules.

A distinguished triangle $U \rightarrow V \rightarrow W \rightarrow U[1]$ is a mapping-cone sequence considered up to quasi-isomorphism. Derived tensor with k gives a long exact sequence containing

$$H_i(k \otimes_A^{\mathbf{L}} U) \longrightarrow H_i(k \otimes_A^{\mathbf{L}} V) \longrightarrow H_i(k \otimes_A^{\mathbf{L}} W).$$

The kernel of the second map has dimension at most $\beta_i(U)$, and its image has dimension at most $\beta_i(W)$. Therefore

$$\beta_i(V) \leq \beta_i(U) + \beta_i(W), \quad \beta^{\text{tot}}(V) \leq \beta^{\text{tot}}(U) + \beta^{\text{tot}}(W).$$

Rotating the triangle gives the corresponding total estimate for either of the other two objects, since shifts do not change total Betti number. We write $H_{\text{tot}}(C) = \bigoplus_i H_i(C)$.

When bases are fixed, the $\mathfrak{m}$ -adic order of a source column indexed by $e \in X_i$ is the largest N for which $d_i(e) \in \mathfrak{m}^N X_{i-1}$; the zero column has order ∞ . We call the column N -deep when its order is at least N , and write $B_N(X)$ for the number of positive-degree source columns that are not N -deep, summed over all differentials. Column depth always means this $\mathfrak{m}$ -adic order. An invertible target-basis change preserves every column's order, because it preserves $\mathfrak{m}^N X_{i-1}$ for every N . A source-basis change can alter the individual column orders.

Remark 2.1 (optimizing the choice of bases). For each fixed $N \geq 1$, the smallest exceptional-column count has an intrinsic description. Put $Q = A/\mathfrak{m}^N$ and let $\mu_Q(I)$ denote the minimal number of generators of a finite Q -module I . Then

$$\min_{\text{choices of bases}} B_N(X) = \sum_{i \geq 0} \mu_Q(\text{im}(d_i \otimes_A Q)).$$

To see this for one differential, write $I = \text{im}(d_i \otimes_A Q)$. The exceptional columns are exactly the columns with nonzero image in I , so their images generate I . There must therefore be at least $\mu_Q(I)$ of them.

Conversely, choose a subset of the column images whose classes form a basis of $I/\mathfrak{m}_Q I$. Nakayama's lemma says that these $\mu_Q(I)$ images generate I . Express every remaining column image as a Q -linear combination of the selected ones. Lift each coefficient to A and subtract that combination of the selected source vectors from the corresponding remaining source vector. These are invertible elementary basis changes. Every remaining column now vanishes modulo $\mathfrak{m}^N$, proving the reverse bound.

The minima can be attained simultaneously, in increasing homological degree. Choosing a source basis for d_i changes only the target basis for d_{i+1} , so it does not constrain the number subsequently attained there. Later source changes leave the already treated differentials unchanged. The image in the formula is taken *after* reducing the differential; it is not in general $(\text{im } d_i) \otimes_A Q$. The optimizing bases may depend on N .

2.2. From finite-length homology to Betti numbers.

Lemma 2.2. *If $(A, \mathfrak{m}, k)$ is regular local of dimension d and V has finite length, then*

$$\beta_i^A(V) \leq \binom{d}{i} \ell_A(V), \quad \sum_i \beta_i^A(V) \leq 2^d \ell_A(V).$$

If C is a bounded complex with finite-length homology, then

$$\beta_A^{\text{tot}}(C) \leq 2^d \sum_j \ell_A H_j(C).$$

Proof. The Koszul complex on a regular system of parameters is a minimal free resolution of k , with rank $\binom{d}{i}$ in degree i . Consequently $\beta_i^A(k) = \binom{d}{i}$. For an exact sequence $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$, the long exact Tor sequence gives $\beta_i(V) \leq \beta_i(U) + \beta_i(W)$. Apply this along a composition series of V , whose $\ell_A(V)$ factors are all isomorphic to k . This proves the first inequality; summing it over i and using $\sum_i \binom{d}{i} = 2^d$ proves the second.

For the complex assertion, let K be the same Koszul resolution of k . The bounded double complex $K_p \otimes_A C_q$ has total complex $K \otimes_A C$, which computes $k \otimes_A^{\mathbf{L}} C$. Here we resolve k instead

of C ; a bounded free resolution can be used in either tensor factor, even when the other complex is not termwise free. Take homology first in the C direction. Since K_p is free, the result is $K_p \otimes_A H_q(C)$. Taking homology in the K direction then gives the spectral sequence

$$E_{p,q}^2 = \mathrm{Tor}_p^A(k, H_q(C)) \implies H_{p+q}(k \otimes_A^{\mathbf{L}} C).$$

There are only finitely many relevant pairs (p, q) : one has $0 \leq p \leq d$, and C is bounded. Thus the spectral sequence converges with a finite filtration on each group on the right. Each associated graded piece is a subquotient of the corresponding E^2 term. In each total degree n this gives

$$\beta_n^A(C) \leq \sum_{p+q=n} \dim_k \mathrm{Tor}_p^A(k, H_q(C)).$$

Summing over n and applying the module estimate to every $H_q(C)$ yields

$$\beta_A^{\mathrm{tot}}(C) \leq \sum_q \sum_p \beta_p^A(H_q(C)) \leq 2^d \sum_q \ell_A H_q(C),$$

as required. $\square$

2.3. Minimization without losing deep columns. Later we attach a small number of free summands to a given complex in order to obtain a resolution. Minimizing that resolution mixes its bases. The next lemma records why the deep columns can still be tracked through those changes.

Lemma 2.3 (unit cancellation with labels). *Let Z be a bounded free complex over a local ring. Label some basis vectors principal, and suppose every principal source column belongs to $\mathfrak{m}Z_{i-1}$. Label a subset of the principal vectors good, and suppose their source columns belong to $\mathfrak{m}^N Z_{i-1}$. If the total number of nonprincipal vectors is e , then Z can be minimized so that every surviving good column is still N -deep, at most e good vectors are deleted, and at most e surviving vectors that were not principal are present.*

Proof. A nonminimal differential has a unit entry. Its source cannot be principal, because every principal column has all entries in $\mathfrak{m}$. Choose a unit pivot in d_i , with nonprincipal source v and target w , and rescale v so that the coefficient of w in $d_i(v)$ is one. Replacing w by $w' = d_i(v)$ is an invertible target-basis change. It preserves all ideal-valued column conditions, and $d_{i-1}(w') = 0$ by the chain identity.

For every other source vector u in degree i , let c_u be the coefficient of w' in $d_i(u)$ in this new target basis, and replace u by $u' = u - c_u v$. If u is principal, then $c_u \in \mathfrak{m}$; if it is good, then $c_u \in \mathfrak{m}^N$. In the latter case, for example,

$$d_i(u') = d_i(u) - c_u d_i(v) \in \mathfrak{m}^N Z_{i-1}.$$

The principal condition is preserved by the same computation with $N = 1$. We carry each label along with its changed basis vector. Changing the basis in degree i is also a target-basis change for d_{i+1} , so the incoming principal and good columns retain their orders.

After these operations $d_i(v) = w'$ and every other source has zero w' -coefficient. The identity $d_i d_{i+1} = 0$ then forces every incoming column to have zero v -coefficient. Together with $d_{i-1}(w') = 0$, this shows that $Av \rightarrow Aw'$ is a direct summand complex, with its only differential an isomorphism. Delete this contractible two-term summand. The remaining complex is quasi-isomorphic to Z .

Each cancellation consumes a nonprincipal source and deletes at most one principal target. It introduces no new basis vectors. There are therefore at most e cancellations, at most e deletions of good vectors, and at most e surviving nonprincipal vectors. Continuing until no unit entry remains gives a minimal complex and proves the assertion. $\square$

Lemma 2.4 (a free comparison with controlled columns). *Let $(A, \mathfrak{m}, k)$ be a Noetherian local ring. Let G be a bounded complex of finite free modules in nonnegative degrees, with chosen bases and $d(G) \subseteq \mathfrak{m}G$. Let M be a finite module of finite projective dimension, and let $f : G \rightarrow M$ be a morphism in the derived category, with M placed in degree zero. Set*

$$C = \text{Cone}(f), \quad \eta = \beta_A^{\text{tot}}(C).$$

For any fixed integer $N \geq 1$, a minimal free resolution P of M has choices of bases satisfying

$$|\beta_i^A(M) - \text{rank}_A G_i| \leq \eta, \quad B_N(P) \leq B_N(G) + \eta. \quad (2.1)$$

More precisely, if E is a minimal free model of C , there is a free resolution Z of M with a degreewise split exact sequence

$$0 \longrightarrow G \longrightarrow Z \longrightarrow E \longrightarrow 0. \quad (2.2)$$

The same assertions hold over a standard graded polynomial ring, with homogeneous bases, modules, and degree-zero morphisms.

Proof. First choose a finite free resolution of M . Since G is bounded free, the derived morphism f can be represented by a chain map from G to that resolution. Its cone is bounded finite free. Minimize this cone to obtain E ; then $\sum_i \text{rank}_A E_i = \eta$.

The cone triangle shows that $H_i(C) = 0$ for $i < 0$, since both G and M have zero homology there. This also forces $E_i = 0$ for $i < 0$. Indeed, if $j < 0$ were the lowest index with $E_j \neq 0$, then

$$H_j(E) = E_j / \text{im}(d_{E,j+1})$$

would be nonzero: minimality puts the denominator in $\mathfrak{m}E_j$, and Nakayama's lemma rules out $\mathfrak{m}E_j = E_j$. If $H_0(f)$ is surjective, then $H_0(C) = 0$ as well, and the same argument gives $E_0 = 0$. We will use this stronger conclusion in the box construction.

The triangle $G \rightarrow M \rightarrow C \rightarrow G[1]$ can now be rotated to $C[-1] \rightarrow G \rightarrow M \rightarrow C$. Replace C by E and represent its first morphism by a chain map $E[-1] \rightarrow G$. A bounded complex of projectives represents its derived morphisms by chain maps, so this replacement is possible. The cone of this map represents M and has the explicit form

$$Z_i = G_i \oplus E_i, \quad d_{Z,i} = \begin{pmatrix} d_{G,i} & \alpha_i \\ 0 & d_{E,i} \end{pmatrix}, \quad \alpha_i : E_i \rightarrow G_{i-1}.$$

The shift by -1 is what makes the added summand in degree i equal to E_i . As both G and E are nonnegative, Z is a nonnegative free resolution of M . Inclusion of the first summand and projection onto the second give (2.2).

Label the G -vectors principal and its N -deep vectors good. The zero block in the displayed differential is essential: each principal source has exactly its old column in G , with no extra entry in an E -target. There are η added vectors. Apply Lemma 2.3 to minimize Z . Every surviving good column remains N -deep. A surviving exceptional column is either one of the original exceptional principal columns or one of the added vectors, so $B_N(P) \leq B_N(G) + \eta$.

For the rank estimate, tensor the split sequence with k and use its homology sequence. Because G and E are minimal, the relevant segment is

$$k \otimes_A E_{i+1} \longrightarrow k \otimes_A G_i \longrightarrow \text{Tor}_i^A(k, M) \longrightarrow k \otimes_A E_i.$$

Writing $e_i = \text{rank}_A E_i$ gives

$$\text{rank}_A G_i - e_{i+1} \leq \beta_i^A(M) \leq \text{rank}_A G_i + e_i.$$

Since $e_i + e_{i+1} \leq \eta$, this implies the first bound in (2.1).

In the graded case, take graded resolutions and homogeneous connecting maps throughout. A homogeneous unit pivot is a nonzero constant between equal shifts. The basis changes in Lemma 2.3

therefore have the required degrees and remain homogeneous. Graded Nakayama supplies the same nonnegative-support argument, and no bound on the internal shifts is needed. $\square$

2.4. A scalar that is nullhomotopic on the error. For a chain endomorphism, being nullhomotopic means that it has the form $dh + hd$ for maps $h_i : C_i \rightarrow C_{i+1}$. The next estimate uses such a homotopy for multiplication by a scalar; annihilation of the homology alone is not the hypothesis.

Lemma 2.5. *Let Q be an Artinian ring, let C be a bounded complex of finite free Q -modules of total rank at most B , and suppose multiplication by $a \in Q$ is nullhomotopic on C . Then*

$$\ell_Q H_{\text{tot}}(C) \leq B \ell_Q(Q/aQ).$$

Proof. Choose h with $dh + hd = a \text{id}_C$. In degree i , the change $(y, x) \mapsto (y - h_{i-1}(x), x)$ identifies the cone of the zero map on C with the cone of multiplication by a . Thus

$$\text{Cone}(a : C \rightarrow C) \cong C \oplus C[1], \quad \ell_Q H_{\text{tot}}(\text{Cone}(a)) = 2\ell_Q H_{\text{tot}}(C).$$

To estimate the left side, filter C by the subcomplexes consisting of its terms in degrees at most j , and give $\text{Cone}(a)$ the induced filtration. The quotient at level j is, up to a homological shift and a harmless differential sign, the two-term complex $C_j \xrightarrow{a} C_j$. If $b_j = \text{rank}_Q C_j$, its homology has total length

$$b_j(\ell_Q(0 :_Q a) + \ell_Q(Q/aQ)) = 2b_j \ell_Q(Q/aQ).$$

The equality follows from the exact sequence $0 \rightarrow (0 :_Q a) \rightarrow Q \xrightarrow{a} Q \rightarrow Q/aQ \rightarrow 0$. For each step of the finite filtration, the long exact homology sequence bounds the total homology length of the middle complex by the sum for the subcomplex and the quotient. Iterating gives

$$\ell_Q H_{\text{tot}}(\text{Cone}(a)) \leq 2 \left(\sum_j b_j \right) \ell_Q(Q/aQ) \leq 2B \ell_Q(Q/aQ).$$

Combining the two estimates and dividing by two proves the lemma. $\square$

3. THE GRADED SPARSE-COLUMN INEQUALITY

Throughout this section fix integers $d, N \geq 1$ and a field k , and let

$$S = k[x_1, \dots, x_d], \quad \mathfrak{m} = (x_1, \dots, x_d),$$

with its standard grading, and let V be a finite-length graded S -module. Write its minimal graded free resolution as

$$F_p = \bigoplus_j S(-j)^{\beta_{p,j}}, \quad \beta_p = \sum_j \beta_{p,j}.$$

Fix homogeneous bases. In F_p , $p \geq 1$, mark B_p source columns exceptional and suppose that every other column of d_p has all its entries in $\mathfrak{m}^N$. Put $B = \sum_{p=1}^d B_p$. Because V is supported at $\mathfrak{m}$, the sheafification of F on $\mathbb{P}_k^{d-1}$ is exact.

The aim is to bound $\ell_k(V)$ using the total Betti numbers and the numbers B_p , without imposing a bound on any graded shift. There are three steps. First, a cut in the generator degrees of a complex over $k[t]$ with torsion homology has a negative Euler characteristic only when a differential crosses the cut. Second, the categorical pairing places the columns of one differential of F in a window of width N ; inside that window, a column whose entries lie in $\mathfrak{m}^N$ cannot cross the cut. This bounds the negative contribution by the exceptional columns. Finally, summing over all translations removes the graded shifts, and positive interpolation isolates any chosen total Betti number.

3.1. A filtered estimate over one variable. Give $k[t]$ its standard grading, with $\deg t = 1$. For a graded free $k[t]$ -module E , let $E^{\leq c}$ be the submodule generated by all homogeneous elements of degrees at most c . This description is intrinsic, so every degree-zero automorphism of E and its inverse preserve $E^{\leq c}$. Explicitly, if $E = \bigoplus_j k[t](-j)^{b_j}$, then

$$E^{\leq c} = \bigoplus_{j \leq c} k[t](-j)^{b_j}, \quad E/E^{\leq c} \cong \bigoplus_{j > c} k[t](-j)^{b_j}.$$

Thus these are cuts by generator shifts, and the displayed quotient is free over $k[t]$. The cut retains an entire free summand once its generator degree is at most c ; it does not discard the higher-degree multiples of that generator. This distinction is necessary for the rank in (3.2) below to detect a differential.

For a bounded complex K of finitely generated graded free $k[t]$ -modules with degree-zero differentials, assume that K is minimal in the sense that $d(K) \subseteq (t)K$, and write $K_h = \bigoplus_j k[t](-j)^{\beta_{h,j}(K)}$. The functional below is a truncated Euler characteristic. When K has torsion homology, its negative contributions come precisely from two-term bars in degrees $1 \rightarrow 0$ whose source and target shifts both lie in $(a, a + N]$. For $a \in \mathbb{Z}$ define

$$\lambda_{a,N}(K) = \sum_{j \leq a} \beta_{0,j}(K) - \sum_{j \leq a+N} \beta_{1,j}(K) + \sum_{h \geq 2, j} (-1)^h \beta_{h,j}(K). \quad (3.1)$$

Let $\rho_{a,N}(K)$ be the rank over $k(t)$ of the induced map

$$K_1^{\leq a+N} \longrightarrow K_0/K_0^{\leq a}. \quad (3.2)$$

The estimate below is a quantitative variation on the graded partial Euler characteristic inequalities of Eisenbud–Erman [7, Definition 4.1 and Proposition 4.2]. Their positivity proof for complexes with torsion homology reduces to shifted two-term resolutions over $k[t]$. We use the same decomposition, but allow the two cutoff degrees to differ by N and retain the rank of the differential between the surviving pieces. This gives the explicit defect needed for our column estimate.

Lemma 3.1. *One has*

$$\lambda_{a,N}(K) \geq -\rho_{a,N}(K) - \sum_{\substack{h > 0 \\ h \text{ odd}}} \text{rank}_{k[t]} H_h(K). \quad (3.3)$$

If all homology of K is $k[t]$ -torsion, the final sum is zero.

Proof. We first put the whole complex into a form in which the inequality can be checked one summand at a time. Every nonzero homogeneous entry of a differential over $k[t]$ is a scalar multiple of a power of t . Choose an entry with smallest exponent in a nonzero differential d_h . It divides all other entries of that matrix. After scaling the entry to t^e , clear its row and column by elementary operations. These operations are homogeneous: the quotient of two entries in a fixed row or column has exactly the degree required for the corresponding change of source or target basis. Repeating the procedure diagonalizes d_h by homogeneous basis changes.

Consider one resulting diagonal entry, from a basis vector x of K_h to a basis vector y of K_{h-1} , with $d_h(x) = t^e y$. Since $d_{h-1}d_h = 0$, we have $t^e d_{h-1}(y) = 0$, and the freeness of K_{h-2} implies $d_{h-1}(y) = 0$. Likewise, $d_h d_{h+1} = 0$ implies that the x -coordinate of every element of $\text{im } d_{h+1}$ is zero: applying d_h would otherwise give a nonzero y -coordinate. Thus this diagonal entry is an isolated two-term direct summand of the complex. Remove all such summands and repeat with a remaining nonzero differential. The total free rank decreases at each removal, so the process ends. It expresses K as a direct sum of two-term bars and one-term free complexes. Minimality ensures that each bar exponent is positive.

This is a simultaneous chain decomposition, rather than separate normal forms that might be incompatible in adjacent degrees. Its basis changes preserve the intrinsic submodules $K_h^{\leq c}$, so they preserve both $\lambda_{a,N}(K)$ and the rank $\rho_{a,N}(K)$. Both quantities are additive on the resulting direct sum.

A bar in positions $1 \rightarrow 0$ has the form

$$k[t](-v) \xrightarrow{t^{v-u}} k[t](-u), \quad v > u.$$

Its contribution to (3.1) is $\mathbf{1}_{u \leq a} - \mathbf{1}_{v \leq a+N}$. This equals -1 exactly when $u > a$ and $v \leq a+N$, precisely when the bar contributes one to the rank in (3.2): both its source and its target survive the indicated cuts, and multiplication by t^{v-u} has rank one over $k(t)$. Every other two-term bar has nonnegative contribution: a bar in positions $2 \rightarrow 1$ contributes $1 - \mathbf{1}_{u \leq a+N}$, a bar in positions $0 \rightarrow -1$ contributes $\mathbf{1}_{v \leq a}$, and bars farther away contribute zero. A one-term free summand can contribute negatively only in a positive odd homological degree, and then by at most one. Such a summand contributes one to the free rank of homology in that same degree; the bars have torsion homology. Adding the contributions gives (3.3) and its asserted torsion specialization. $\square$

3.2. Translated supernatural tests. We recall only the part of the Eisenbud–Erman pairing needed below. For a bounded graded free S -complex F and a vector bundle $\mathcal{E}$ on $\mathbb{P}_k^{d-1}$, their construction gives an object $\Phi(F, \mathcal{E})$ of $D^b(\text{gr } k[t])$ [7, Definition 2.1]. Choose a bounded minimal graded free representative and use the same notation for it. When F is minimal, with $F_p = \bigoplus_j S(-j)^{\beta_{p,j}}$, the free summands of $\Phi(F, \mathcal{E})$ are obtained from

$$H^q(\mathbb{P}^{d-1}, \mathcal{E}(-j))^{\beta_{p,j}} \otimes_k k[t](-j) \quad (3.4)$$

in homological degree $p - q$. Theorem 2.3 of [7] gives this Betti table. For the sparse-column argument, the table alone is not enough: we must identify which original column can contribute to a specific entry of the paired differential. The next lemma records the needed representative and the labels that retain this information. Here a contraction means an inclusion of chosen cohomology representatives, a projection onto them, and a homotopy from their composite to the identity; these maps are constructed in the proof.

Lemma 3.2 (columnwise form of the pairing). *Assume that F is minimal and that its sheafification is exact. Choose a contraction of the vertical Čech complex of each twist onto its cohomology, and use that contraction on each original basis line with that twist. Then $\Phi(F, \mathcal{E})$ has a minimal representative such that:*

(1) *its differential is*

$$d_{\text{tr}} = \sum_{s \geq 0} \epsilon_s \pi \delta (h \delta)^s \iota, \quad \epsilon_s \in \{1, -1\}, \quad (3.5)$$

where δ is the horizontal differential of F , h is the chosen vertical contraction, and the convention-dependent signs ϵ_s are immaterial;

- (2) *each generator in (3.4) retains the label of its original basis vector of F , together with a basis label in the corresponding cohomology group;*
- (3) *every nonzero transferred differential has positive t -degree, so the resulting $k[t]$ -complex is minimal;*
- (4) *its homology is $k[t]$ -torsion.*

Proof. The pairing is the derived direct image to $\text{Spec } k[t]$ of the double complex obtained after the substitution $x_j \mapsto tx_j$ and tensoring with $\mathcal{E}$. Compute it by the Čech complex for the standard affine cover of projective space. For one original basis line $S(-j)$ in F_p , its vertical complex is the Čech complex $C^\bullet(\mathcal{E}(-j))$, tensored with $k[t](-j)$. Here the horizontal index is p and the vertical cohomological index is q ; total homological degree is $p - q$.

We describe the contraction over k before extending scalars. In each C^q , choose representatives H^q for cohomology, the subspace B^q of boundaries, and a complement U^q to the cycles. Thus $C^q = B^q \oplus H^q \oplus U^q$, and the vertical differential restricts to an isomorphism $U^q \rightarrow B^{q+1}$. Let ι include the chosen H^q , let π project onto it, and let h be the inverse of this isomorphism on B^{q+1} and zero on the other two summands. These maps satisfy

$$\pi\iota = 1, \quad 1 - \iota\pi = d_v h + h d_v.$$

The construction also gives $h^2 = \pi h = h\iota = 0$. Incorporate the usual total-complex signs in these maps. The vertical complexes need not have finite-dimensional terms, but their cohomology is finite-dimensional. Extend the splittings over $k[t]$ and tensor them with the identity on the multiplicity space of each original twist. In particular, vertical maps preserve both the original F -degree and its basis labels. Since δ lowers the bounded F -degree, $h\delta$ is nilpotent. The basic perturbation lemma [12] now gives the finite sum (3.5) and a homotopy equivalence from this transferred complex to the original total complex. Concretely, the first term $\pi\delta\iota$ is the map induced by the horizontal differential on vertical cohomology. A higher term allows the image to pass through a contractible vertical summand, applies h to lift it back one vertical degree, and then applies another horizontal differential. Nilpotence makes this correction process finite, so no convergence or finiteness assumption on the individual Čech terms is needed. The underlying free modules of the transferred complex are exactly (3.4). The labels record these summands; they do not assert that the transferred differential preserves an original basis line.

The index bookkeeping is explicit: a term with $s + 1$ horizontal factors sends original indices (p, q) to $(p - s - 1, q - s)$. Here h lowers the Čech cohomological index by one and preserves p , while δ lowers p and preserves q . Thus the transferred map has total homological degree -1 .

Every summand of (3.5) contains at least one horizontal factor. A nonzero entry $S(-v) \rightarrow S(-u)$ of δ has $v - u \geq 1$ by minimality and, after substitution, contributes the factor t^{v-u} . The vertical maps have t -degree zero. Along a path from shift v to shift u , the exponents telescope to $v - u \geq s + 1$. Hence every transferred differential lies in (t) , proving minimality.

Finally, invert t . Rescaling a basis line of shift j by t^{-j} cancels every factor t^{v-u} in the substituted horizontal differential, leaving the original sheaf complex tensored with $\mathcal{E}$ and $k[t, t^{-1}]$. Because $\mathcal{E}$ is locally free, this sheaf complex is exact. Its sections on every affine intersection in the standard cover are exact as well. Taking horizontal homology first in the bounded Čech double complex therefore shows that its total complex is acyclic. Thus the finite free transferred model becomes acyclic after inverting t , and its finitely generated homology is killed by a power of t . This is the generic-exactness property of the pairing recorded in [7, Proposition 2.6]; it gives the required torsion assertion. This proves all four assertions. $\square$

Assume now that $d \geq 2$. The choice of supernatural roots by omitting two adjacent indexed nodes follows the construction in the proof of Eisenbud–Schreyer [8, Theorem 7.1]. Here we apply that choice to a translated arithmetic progression with arbitrary spacing N . The resulting window will let us retain the original source-column labels and bound the total rank defect over all translations. For $0 \leq i \leq d - 1$ and $a \in \mathbb{Z}$, put

$$b_\ell(a) = a + (\ell - i)N, \quad \ell \in \{0, \dots, d\} \setminus \{i, i + 1\}. \quad (3.6)$$

Choose a supernatural bundle $\mathcal{E}_{i,a}$ with actual root sequence $(-b_\ell(a))_\ell$, in decreasing order. Thus the numbers $b_\ell(a)$ in (3.6) are the negatives of the roots in the convention of [8, Theorem 6.4]. The product construction of Eisenbud–Schreyer, applied with the $d - 1$ roots as singleton consecutive blocks, gives such a bundle over every field with rank $(d - 1)!$ [8, Theorem 6.1]. Away from the roots there is one nonzero cohomology group, and the rank normalization above gives

$$h^q(\mathcal{E}_{i,a}(-j)) = \frac{\text{rank } \mathcal{E}_{i,a}}{(d - 1)!} \prod_\ell |j - b_\ell(a)| = \prod_\ell |j - b_\ell(a)| \quad (3.7)$$

for its unique degree q [8, Theorem 6.4]. The cohomological degree of this group is

$$q = \#\{\ell \in \{0, \dots, d\} \setminus \{i, i+1\} : b_\ell(a) < j\}.$$

At an included root all cohomology vanishes. If $a < j \leq a+N$, exactly the i indices $\ell = 0, \dots, i-1$ satisfy the displayed inequality, while every index $\ell \geq i+2$ fails it. The two omitted indices correspond to a and $a+N$ themselves. Consequently the whole half-open window $(a, a+N]$, including its right endpoint, has cohomology only in degree i . This is the reason for omitting two consecutive roots.

Let $K_{i,a}$ be the specific minimal transferred representative of $\Phi(F, \mathcal{E}_{i,a})$ furnished by Lemma 3.2, and put

$$L_i(a) = \lambda_{a,N}(K_{i,a}).$$

Lemma 3.3. *For every i ,*

$$\sum_{a \in \mathbb{Z}} L_i(a) \geq -d^{d-1} N^d B_{i+1}. \quad (3.8)$$

The sum is finite.

Proof. By Lemma 3.2, $K_{i,a}$ is minimal with torsion homology, so Lemma 3.1 applies without a free-homology error. A component of the restricted map

$$(K_{i,a})_1^{\leq a+N} \longrightarrow (K_{i,a})_0 / (K_{i,a})_0^{\leq a}$$

has target and source shifts u, v with $a < u < v \leq a+N$. Both twists in this interval have cohomology only in degree i . Hence the source comes from F_{i+1} and the target from F_i . A term of (3.5) containing $s+1$ horizontal factors lowers the original F -degree by $s+1$, because the intervening vertical maps preserve that degree. A path from the relevant F_{i+1} -copy to the F_i -copy must lower it by exactly one. Thus only the $s=0$ term contributes, and this term is the map on cohomology induced by d_{i+1} . If its original source column is not exceptional, it is zero. Indeed, every entry of that column is a homogeneous polynomial of degree at least N if it is nonzero, whereas an entry from shift v to shift u in this window would have degree $v-u < N$. There is therefore no such original entry, and applying π and ι cannot produce one.

This rank is estimated in the labeled transferred model. The bar decomposition used in Lemma 3.1 may change the labels, but it preserves the intrinsic cuts and therefore the rank being estimated.

It follows that the rank is at most the sum, over exceptional columns of F_{i+1} with shift $v \in (a, a+N]$, of $h^i(\mathcal{E}_{i,a}(-v))$. A fixed exceptional column occurs for exactly the N integers $a = v-N, \dots, v-1$. Writing $s = v-a$, its weight is

$$\prod_{\ell=1}^i (s + \ell N) \prod_{\ell=2}^{d-i} (\ell N - s).$$

There are $d-1$ factors and each has absolute value at most dN . Thus its total contribution over all translations is at most $N(dN)^{d-1} = d^{d-1} N^d$. Summing this bound over the B_{i+1} exceptional columns bounds the total of the restricted ranks. Lemma 3.1 then gives (3.8). The left side is a finite sum, as the independent formula and moment calculation in Lemma 3.4 below show. $\square$

3.3. Moments and positive interpolation. We now convert the translated estimates into an expression involving only total Betti numbers. The cancellation that removes the shifts comes from the Hilbert-series identity

$$\sum_{p,j} (-1)^p \beta_{p,j} z^j = (1-z)^d \text{Hilb}_V(z)$$

and the fact that $\text{Hilb}_V(z)$ is a finite Laurent polynomial with $\text{Hilb}_V(1) = \ell_k(V)$. Apply the operator $z(d/dz)$ repeatedly to the identity and evaluate at $z = 1$. Its m -th power sends z^j to $j^m z^j$. On the right, all evaluations vanish when $m < d$ because of the factor $(1 - z)^d$. When $m = d$, only the term differentiating that factor d times survives, and it gives $(-1)^d d! \text{Hilb}_V(1)$. This proves the moment identities

$$\sum_{p,j} (-1)^p \beta_{p,j} j^m = 0 \quad (0 \leq m < d), \quad (3.9)$$

$$\sum_{p,j} (-1)^p \beta_{p,j} j^d = (-1)^d d! \ell_k(V). \quad (3.10)$$

These are the Peskine–Szpiro multiplicity equations [24]; their pure-resolution specialization gives the Herzog–Kühl equations [14]. The displayed Hilbert-series calculation fixes our sign and normalization.

For $0 \leq i \leq d - 1$, define

$$Q_i(u) = \prod_{\ell \in \{0, \dots, d\} \setminus \{i, i+1\}} (\ell N - u). \quad (3.11)$$

Lemma 3.4. *One has*

$$L_i(a) = \sum_{p,j} (-1)^p \beta_{p,j} Q_i(j - a + iN) \mathbf{1}_{j-a+iN \leq pN}. \quad (3.12)$$

If $P_i \in \mathbb{Q}[u]$ satisfies $P_i(u) - P_i(u - 1) = Q_i(u)$, then

$$T_i := \sum_a L_i(a) = (d - 1)! \ell_k(V) + \sum_p (-1)^p \beta_p P_i(pN). \quad (3.13)$$

Proof. If $V = 0$, both identities are immediate. Assume $V \neq 0$. Put $u = j - a + iN$ and, away from the included roots, set

$$q_i(u) = \#\{\ell \in \{0, \dots, d\} \setminus \{i, i+1\} : \ell N < u\}.$$

This is the unique cohomology degree of $\mathcal{E}_{i,a}(-j)$, and

$$\text{sgn } Q_i(u) = (-1)^{q_i(u)}.$$

A row- p term of F therefore lies in paired homological degree $h = p - q_i(u)$. By (3.1), the term is retained unconditionally if $h \geq 2$, retained if $u \leq (i+1)N$ when $h = 1$, retained if $u \leq iN$ when $h = 0$, and omitted if $h < 0$. We verify that these conditions amount to the single inequality $u \leq pN$. This is the step that relates the cohomological window to the original resolution row.

- (1) If $p < i$, the node pN is an included root. Below this node there are at most p roots below u , so $h \geq 0$. Moreover $u < pN \leq iN$, so either of the low-degree cuts is satisfied. Above this node there are at least $p + 1$ roots below u , so $h < 0$ and the term is omitted.
- (2) If $p = i$, then $u \leq iN$ implies $q_i(u) \leq i$ and $h \geq 0$, with both low-degree cuts satisfied. When $u > iN$, we have $q_i(u) \geq i$: the only possible nonnegative degree is $h = 0$, and its cut fails.
- (3) If $p = i + 1$, then $u \leq (i+1)N$ implies $q_i(u) \leq i$ and $h \geq 1$. A degree-one term satisfies its cut, and a higher-degree term is retained automatically. When $u > (i+1)N$, we have $h \leq 1$; neither low-degree cut can hold, so the term is omitted.
- (4) If $p > i + 1$, the node pN is an included root. Below it, at most $p - 2$ included roots lie below u , since both omitted nodes precede pN . Thus $h \geq 2$. Above it, at least $p - 1$ included roots lie below u , so $h \leq 1$; both low-degree cuts fail because $u > pN > (i+1)N$.

At an included root the cohomology and $Q_i(u)$ both vanish, so no choice of strict or weak inequality affects its contribution. The two omitted roots are covered by the weak inequalities in the middle two cases. Finally, the signed contribution of every retained nonroot term is

$$(-1)^{p-q_i(u)}|Q_i(u)| = (-1)^p Q_i(u).$$

This proves (3.12).

Only finitely many pairs (p, j) have $\beta_{p,j} \neq 0$. For $a \ll 0$ all their indicators vanish. For $a \gg 0$ they are all one, and the right side is zero: expand $Q_i(j - a + iN)$ as a polynomial in j of degree $d - 1$ and apply (3.9) to each coefficient. Thus $a \mapsto L_i(a)$ has finite support.

A discrete primitive P_i exists because taking $P(u) - P(u - 1)$ lowers the degree of a nonconstant polynomial by one. Since Q_i has leading coefficient $(-1)^{d-1}$, the primitive has degree d and leading coefficient $(-1)^{d-1}/d$. Choose an integer A beyond the finite support of $a \mapsto L_i(a)$ and with

$$A \geq \max_{\beta_{p,j} \neq 0} \{j + iN - pN\}.$$

Then every individual nonzero Betti summand has entered its truncated sum by $a = A$. For a fixed pair (p, j) , the indicator requires $a \geq j + iN - pN$. As a runs from this lower limit to A , the argument $j - a + iN$ runs through the integers from pN down to $j - A + iN$. Consequently telescoping gives

$$\sum_{a \leq A} Q_i(j - a + iN) \mathbf{1}_{j-a+iN \leq pN} = P_i(pN) - P_i(j - A + iN - 1).$$

Multiply by $(-1)^p \beta_{p,j}$ and sum over the finitely many pairs (p, j) . The first term becomes $\sum_p (-1)^p \beta_p P_i(pN)$ because it is independent of j . In the second term, expand $P_i(j - A + iN - 1)$ in powers of j . All powers below d cancel by (3.9); its degree- d coefficient is unchanged by translation. Including the minus sign from telescoping, the surviving term is

$$-\frac{(-1)^{d-1}}{d} \sum_{p,j} (-1)^p \beta_{p,j} j^d = (d-1)! \ell_k(V).$$

This proves (3.13). □

Lemma 3.5 (positive interpolation). *For each $0 \leq r \leq d$ there are nonnegative rational numbers $w_0, \dots, w_{d-1}$ such that*

$$(d-1)! \sum_i w_i = 1 \tag{3.14}$$

and

$$\sum_i w_i T_i = \ell_k(V) - \frac{N^d}{\binom{d}{r}} \beta_r. \tag{3.15}$$

Proof. We seek a nonnegative combination of the T_i whose Betti-row term vanishes at every interpolation node pN except the chosen node rN . Use the following scaled Lagrange polynomial for that node:

$$C_r(u) = -\frac{1}{d!} \prod_{\substack{0 \leq j \leq d \\ j \neq r}} (jN - u), \quad Q(u) = C_r(u) - C_r(u - 1).$$

At the interpolation nodes,

$$Q_i(pN) = 0 \quad \text{unless } i = p - 1 \text{ or } i = p,$$

and, on putting $A_p = (-1)^p p! (d - p)! N^{d-1}$,

$$Q_p(pN) = A_p \quad (0 \leq p < d), \quad Q_{p-1}(pN) = -A_p \quad (0 < p \leq d).$$

These evaluations prove linear independence without a determinant calculation. In a relation $\sum_i c_i Q_i = 0$, evaluation at 0 gives $c_0 A_0 = 0$, hence $c_0 = 0$. Evaluation at pN , for $1 \leq p < d$, gives $A_p(c_p - c_{p-1}) = 0$, so successively every c_p is zero. There are d polynomials, all of degree at most $d - 1$, so they form a basis of that polynomial space.

The polynomial Q has degree $d - 1$. Expand it uniquely as $Q = \sum_i w_i Q_i$, with $w_i \in \mathbb{Q}$, and set $w_{-1} = w_d = 0$. Evaluation at pN , for $0 \leq p \leq d$, gives

$$w_p - w_{p-1} = \frac{(-1)^p (C_r(pN) - C_r(pN - 1))}{p!(d-p)!N^{d-1}}. \quad (3.16)$$

For $p < r$, the right side is nonnegative; for $p > r$, it is nonpositive. To see this, note that $C_r(pN) = 0$ when $p \neq r$. At $u = pN - 1$, the product defining C_r has p negative factors if $p < r$, and $p - 1$ if $p > r$, unless a factor vanishes. Including the initial minus sign in C_r gives the claimed signs in (3.16). A vanishing factor can occur when $N = 1$, and then the increment is zero. Write the right side of (3.16) as Δ_p . For $i < r$, summing from the left gives $w_i = \sum_{p=0}^i \Delta_p \geq 0$. The endpoint equations give $\sum_{p=0}^d \Delta_p = w_d - w_{-1} = 0$; hence, for $i \geq r$, summing from the right gives $w_i = -\sum_{p=i+1}^d \Delta_p \geq 0$. Thus every w_i is nonnegative. The sign of the single increment Δ_r is unnecessary: each weight can be summed from the endpoint on the same side of the chosen node rN .

The leading coefficients are

$$\text{lc}(Q) = \frac{(-1)^{d-1}}{(d-1)!}, \quad \text{lc}(Q_i) = (-1)^{d-1}.$$

Comparing them in $Q = \sum_i w_i Q_i$ gives (3.14). For the primitives in Lemma 3.4, the polynomial $\sum_i w_i P_i - C_r$ has zero discrete difference, because $\sum_i w_i Q_i = Q$. It is therefore a constant. Its contribution to the alternating Betti sum is that constant times $\sum_p (-1)^p \beta_p$, which is zero by the zeroth moment identity. Thus we may replace $\sum_i w_i P_i$ by C_r in that sum. Since

$$C_r(pN) = 0 \quad (p \neq r), \quad C_r(rN) = -(-1)^r \frac{N^d}{\binom{d}{r}},$$

formula (3.13) now gives

$$\begin{aligned} \sum_i w_i T_i &= (d-1)! \left(\sum_i w_i \right) \ell_k(V) + \sum_p (-1)^p \beta_p C_r(pN) \\ &= \ell_k(V) - \frac{N^d}{\binom{d}{r}} \beta_r. \end{aligned}$$

This is (3.15). □

Example (the interpolation weights in dimension two). Take $d = 2$ and choose the row $r = 0$. The two test polynomials and the scaled Lagrange polynomial are

$$Q_0(u) = 2N - u, \quad Q_1(u) = -u, \quad C_0(u) = -\frac{1}{2}(N - u)(2N - u).$$

Taking the discrete difference gives

$$C_0(u) - C_0(u-1) = \frac{3N+1}{2} - u.$$

Thus the equation $w_0 Q_0 + w_1 Q_1 = C_0(u) - C_0(u-1)$ says $w_0 + w_1 = 1$ and $2Nw_0 = (3N+1)/2$. Solving gives

$$w_0 = \frac{3N+1}{4N}, \quad w_1 = \frac{N-1}{4N}.$$

These weights are nonnegative for every $N \geq 1$ and give

$$w_0 T_0 + w_1 T_1 = \ell_k(V) - N^2 \beta_0.$$

At $N = 1$, the second weight vanishes and the first is one. This shows concretely why the endpoint case permits zero weights rather than requiring strict positivity.

Theorem 3.6 (graded sparse-column inequality). *With the notation at the beginning of this section, for every $0 \leq r \leq d$,*

$$\ell_k(V) \geq \frac{N^d}{\binom{d}{r}} \beta_r - C_d N^d B, \quad C_d = \frac{d^{d-1}}{(d-1)!}. \quad (3.17)$$

Proof. Suppose first that $d \geq 2$. Combine Lemmas 3.3 and 3.5:

$$\begin{aligned} \ell_k(V) - \frac{N^d}{\binom{d}{r}} \beta_r &= \sum_i w_i T_i \\ &\geq -d^{d-1} N^d \sum_i w_i B_{i+1} \\ &\geq -\frac{d^{d-1}}{(d-1)!} N^d B. \end{aligned}$$

For $d = 1$, finite length gives $\text{rank } F_0 = \text{rank } F_1$, so the presentation matrix is square; its graded invariant factors over the PID $k[x]$ are powers x^{e_ℓ} , with $e_\ell \geq 1$ by minimality. There are $\beta_0 = \beta_1$ such factors, and $\ell_k(V) = \sum_\ell e_\ell$.

Reduce the presentation map modulo x^N and put $A = k[x]/(x^N)$. Its nonexceptional columns become zero, so its image as an A -module is generated by at most B elements. Smith form identifies this image with the direct sum of the nonzero ideals $x^{e_\ell} A$ for which $e_\ell < N$. Each of these ideals has one-dimensional quotient by its maximal-ideal multiple, so their number is the minimal number of generators of the image. It is therefore at most B . All the other invariant factors have $e_\ell \geq N$, and hence

$$\ell_k(V) = \sum_\ell e_\ell \geq N(\beta_0 - B).$$

This is the required inequality for both rows, since $C_1 = 1$. □

Corollary 3.7. *Under the same hypotheses,*

$$\ell_k(V) \geq (N/2)^d \sum_{r=0}^d \beta_r - C_d N^d B.$$

Proof. Multiply the row- r inequality (3.17) by $\binom{d}{r}/2^d$ and sum over r . These weights sum to one, so the left side and the common error term retain their coefficients. The Betti term in each row becomes $(N/2)^d \beta_r$, giving the displayed inequality. □

Remark 3.8. The main term is sharp. For $V = S/(x_1^N, \dots, x_d^N)$, the Koszul resolution has no exceptional columns, $\ell_k(V) = N^d$, and $\beta_r = \binom{d}{r}$, so equality holds in every row of (3.17). For $N = 1$ the row inequality is the elementary estimate $\beta_r(V) \leq \binom{d}{r} \ell_k(V)$ from Lemma 2.2. We do not assert that the coefficient of B is optimal.

This completes the graded part of the argument. The next two sections transport both the inequality and its chosen deep columns to an arbitrary equicharacteristic regular local ring.

4. PRESCRIBED-JET GLOBALIZATION

Theorem 3.6 applies to graded resolutions. A local resolution need not admit a compatible grading, and its associated graded complex need not be exact. We therefore make the passage from local to graded in two stages. First, we replace the local resolution by an exact polynomial resolution, retaining its ranks and the chosen deep columns after taking a finite direct sum. Then, in Section 5, we replace this polynomial resolution by graded resolutions with the required asymptotic invariants.

The first stage is possible because a finite-length module over a power-series ring is already defined over a polynomial quotient. The difficulty is to choose polynomial generators for its successive syzygies while preserving the desired columns. We preserve their images modulo a fixed power of the maximal ideal; these finite approximations are the *jets* in the section title.

The generator argument below is a prescribed-jet variant of the dimension-counting method underlying the Forster–Swan theorem [11, 30]. We give the argument because its approximation property, rather than a generator bound alone, is needed here.

Throughout this section let $d \geq 1$, $S = k[x_1, \dots, x_d]$, $\mathfrak{m} = (x_1, \dots, x_d)$, and $A = k[[x_1, \dots, x_d]]$, with maximal ideal $\mathfrak{m}_A = \mathfrak{m}A$.

Lemma 4.1 (prescribed generators). *Assume that k is infinite. Let K be a finite S -module that is locally free of rank $r \geq 1$ on $X = \operatorname{Spec} S \setminus \{\mathfrak{m}\}$. Suppose $v_1, \dots, v_s \in K$ generate $K_{\mathfrak{m}}$ and $s \geq r + d$. For every $N \geq 1$, there are global generators $v'_1, \dots, v'_s$ of K with*

$$v'_j - v_j \in \mathfrak{m}^N K \quad (1 \leq j \leq s).$$

Proof. Choose global generators $w_1, \dots, w_t$ of $\mathfrak{m}^N K$. We seek the new generators among the vectors

$$v_j(c) = v_j + \sum_{\ell=1}^t c_{\ell j} w_{\ell}, \quad 1 \leq j \leq s,$$

where the ts coefficients $c_{\ell j}$ are independent parameters. Every choice has the prescribed jets. Moreover, every choice still generates at $\mathfrak{m}$: the perturbations lie in $\mathfrak{m}K_{\mathfrak{m}}$, so the images in $K_{\mathfrak{m}}/\mathfrak{m}K_{\mathfrak{m}}$ are unchanged. Nakayama's lemma then applies. We need one choice that also generates everywhere on X .

At a geometric point $x \in X$, some element of $\mathfrak{m}$ is invertible. Thus $\mathfrak{m}^N \mathcal{O}_{X,x} = \mathcal{O}_{X,x}$, and the images of the w_{ℓ} span the r -dimensional fiber of K . Consequently, after choosing a frame for that fiber, the affine map

$$\mathbb{A}^{ts} \longrightarrow \operatorname{Mat}_{r \times s}, \quad c \longmapsto (v_1(c)(x), \dots, v_s(c)(x))$$

is surjective. The columns fail to generate precisely when this matrix has rank less than r .

The determinantal locus of $r \times s$ matrices of rank at most $r - 1$ has codimension $s - r + 1$: its largest stratum has dimension $(r - 1)(s + 1)$ inside the rs -dimensional matrix space. Hence, for each x , the bad parameters have codimension $s - r + 1$. Apply this fiberwise calculation to the incidence of pairs (x, c) for which generation fails. Local trivializations of the vector bundle $K|_X$ show that this incidence has dimension at most

$$\dim X + ts - (s - r + 1) \leq ts + d - (s - r + 1) \leq ts - 1.$$

Its image in the ts -dimensional parameter space therefore has proper Zariski closure. The complement is a nonempty open subset of $\mathbb{A}_k^{ts}$, and it has a k -point because k is infinite. Choose such a point. The resulting vectors generate every fiber on X , hence generate at every local ring on X by Nakayama. They already generate at $\mathfrak{m}$. The cokernel of the resulting map $S^s \rightarrow K$ thus localizes to zero at every prime, so it is zero. $\square$

At the final syzygy the number of available vectors equals the rank, so the preceding generator surplus is no longer available. We need a polynomial basis approximating a formal basis. There is one obstruction to an arbitrary such approximation: a polynomial invertible matrix has constant determinant. The next lemma removes that obstruction by rescaling one formal basis vector. This is harmless for our application, since multiplying a column by a unit does not change its depth.

Lemma 4.2 (basis jets). *Let $K \cong S^r$, $r > 0$, let $N \geq 1$, and let $v_1, \dots, v_r$ be an A -basis of $A \otimes_S K$. After multiplying at most one v_j by a unit of A and retaining the notation v_j for the resulting vectors, there is an S -basis $v'_1, \dots, v'_r$ such that*

$$v'_j - v_j \in \mathfrak{m}_A^N(A \otimes_S K).$$

Proof. Fix an S -basis e of K , and write the formal basis as $v = eU$ with $U \in \mathrm{GL}_r(A)$. Let $d_U = \det U$ and let $d_0 = d_U(0) \in k^\times$. Multiplying the first column of U by the unit d_0/d_U changes its determinant to d_0 . We make this normalization first.

The quotient rings $A/\mathfrak{m}_A^N$ and $S/\mathfrak{m}^N$ are canonically equal. Thus the reduction of the normalized matrix can be viewed as an invertible matrix over $S/\mathfrak{m}^N$. After multiplying by a constant diagonal matrix of determinant d_0^{-1} , it belongs to $\mathrm{SL}_r(S/\mathfrak{m}^N)$. For $r \geq 2$, every special linear matrix over this local ring is a product of elementary matrices. Indeed, elimination with unit pivots reduces it to determinant-one diagonal blocks, and such blocks are elementary because

$$\begin{pmatrix} u & 0 \\ 0 & u^{-1} \end{pmatrix} = w(u)w(-1), \quad w(u) = \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -u^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}.$$

Lift the coefficient of each elementary matrix from $S/\mathfrak{m}^N$ to S . Each lifted elementary matrix is invertible over S , so their product is an invertible polynomial lift. Restore the constant diagonal matrix used above. The columns of the resulting matrix give the required S -basis and have the prescribed reductions modulo $\mathfrak{m}_A^N$. If $r = 1$, the initial normalization already makes the single basis coefficient constant, and the conclusion follows directly. $\square$

Proposition 4.3 (globalization with selected columns). *Let $M \neq 0$ be a finite-length A -module with minimal resolution F . In every positive degree i , mark g_i basis vectors whose columns are N -deep, where $N \geq 1$, and put*

$$B = \sum_{i=1}^d (\beta_i(F) - g_i).$$

If k is infinite, then for every integer $a \geq d$ there is an exact polynomial free resolution P of $M^{\oplus a}$, supported in degrees $0, \dots, d$ and minimal at $\mathfrak{m}$, with

$$\mathrm{rank}_S P_i = a\beta_i(F)$$

and at least ag_i marked N -deep columns in degree i . Here M is viewed as an S -module through $S \rightarrow A$. If k is finite, the same conclusion holds for some positive integer a .

Proof. Choose C with $\mathfrak{m}_A^C M = 0$. Since $S/\mathfrak{m}^C = A/\mathfrak{m}_A^C$, the action on M factors through the same finite-dimensional quotient of either ring. In particular, M is a finite S -module supported at $\mathfrak{m}$, and $A \otimes_S M \cong M$.

Assume first that k is infinite. We explain why taking a copies provides exactly the generator surplus needed in the preceding lemma. For $1 \leq i \leq d$, set $r_i = \mathrm{rank}_A \mathrm{im}(F_i \rightarrow F_{i-1})$, and put $r_0 = r_{d+1} = 0$. A nonzero finite-length module over the d -dimensional regular local ring A has projective dimension d , by the Auslander–Buchsbaum formula. Its minimal resolution has no zero term between F_0 and F_d : exactness and Nakayama’s lemma would otherwise force all subsequent terms to vanish. Similarly, $\mathrm{im}(F_i \rightarrow F_{i-1})$ cannot vanish for $1 \leq i \leq d$, since then minimality and exactness would force $F_i = 0$. Each of these images is a nonzero submodule of a free module over

the domain A , so it has positive rank. Tensoring the resolution with the fraction field, or taking ranks in its syzygy sequences, gives

$$\beta_i(F) = r_i + r_{i+1}, \quad r_i > 0 \quad (1 \leq i \leq d). \quad (4.1)$$

The equality also holds for $i = 0$, since M has rank zero.

Start with a surjection $P_0 = S^{a\beta_0} \rightarrow M^{\oplus a}$ given by the specified minimal generators. Suppose that the exact polynomial complex has been constructed through P_{i-1} , where $1 \leq i < d$, and that its localization at $\mathfrak{m}$ is minimal. Write K_i for the next kernel. For $i = 1$ this is the kernel of the augmentation; for $i > 1$ it is the kernel of $P_{i-1} \rightarrow P_{i-2}$.

There are two facts about K_i to check. First, on the punctured spectrum the resolved module is zero. The successive exact sequences there are locally split: starting at degree zero, their quotients are locally free, and the assertion follows inductively. Thus K_i is locally free on the punctured spectrum. Second, completion identifies the constructed minimal partial resolution with the corresponding part of $F^{\oplus a}$, by uniqueness of minimal free resolutions. It follows that K_i has rank ar_i and that $(K_i)_{\mathfrak{m}}$ needs $a\beta_i$ generators.

Choose such a completed comparison. The images of the given basis of $F_i^{\oplus a}$ then provide a minimal generating set $z_1, \dots, z_{a\beta_i}$ of $A \otimes_S K_i$. Lift their classes modulo $\mathfrak{m}_A^N$ to elements $v_j \in K_i$, using the canonical isomorphism

$$K_i / \mathfrak{m}^N K_i \cong (A \otimes_S K_i) / \mathfrak{m}_A^N (A \otimes_S K_i).$$

Their images modulo $\mathfrak{m}$ form the same generating set as the z_j , so the v_j generate $(K_i)_{\mathfrak{m}}$ by Nakayama's lemma.

We record separately why the marked columns survive this operation. A marked original image is in $\mathfrak{m}_A^N F_{i-1}^{\oplus a}$. Its transported image z_j is therefore in $\mathfrak{m}_A^N (A \otimes_S P_{i-1})$, because an invertible target map preserves this submodule. The lifting error belongs to

$$\mathfrak{m}_A^N (A \otimes_S K_i) \subseteq \mathfrak{m}_A^N (A \otimes_S P_{i-1}).$$

Hence v_j also belongs to the latter submodule after completion. Since P_{i-1} is free and $S \cap \mathfrak{m}_A^N = \mathfrak{m}^N$, this means $v_j \in \mathfrak{m}^N P_{i-1}$ before completion.

The surplus required by Lemma 4.1 is

$$a\beta_i - ar_i = ar_{i+1} \geq a \geq d.$$

Apply that lemma to replace the v_j by global generators of K_i , changing each by an element of $\mathfrak{m}^N K_i$. These changes preserve the marked depth, because $\mathfrak{m}^N K_i \subseteq \mathfrak{m}^N P_{i-1}$. Use the resulting generators as the columns of $P_i = S^{a\beta_i} \rightarrow P_{i-1}$. Its image is exactly K_i , so the complex remains exact. At $\mathfrak{m}$ the generators are still minimal, as their classes modulo $\mathfrak{m} K_i$ are unchanged.

This construction is repeated degree by degree. A comparison chosen at a later stage need not preserve the earlier chosen frames, but it is used only to transport generators of the next kernel. It does not change the polynomial maps already constructed. The invariant being preserved throughout is the depth of each marked column, not simultaneous equality of all differential matrices with their original jets.

After P_{d-1} has been constructed, the last kernel K_d requires a different argument. Its completion is the final free syzygy of $F^{\oplus a}$, of rank $a\beta_d$. Freeness descends along the faithfully flat completion $S_{\mathfrak{m}} \rightarrow A$, so $(K_d)_{\mathfrak{m}}$ is free. At every other prime it is locally free by the preceding splitting argument. Thus K_d is a finite projective S -module. The Quillen–Suslin theorem makes it free [25, 29]. Apply Lemma 4.2 to the completed basis transported from $F_d^{\oplus a}$. The possible rescaling by a unit preserves the depth of the rescaled column, and the same lifting argument as above preserves every marked depth. The resulting polynomial basis gives $P_d \cong K_d$ and completes the exact resolution. Its localization at $\mathfrak{m}$ is minimal, as can also be checked after faithfully flat completion.

If k is finite, extend the coefficients to an algebraic closure and perform this construction with $a = d$. Since M is annihilated by $\mathfrak{m}^C$, its coefficient extension is defined using the finite quotient $S/\mathfrak{m}^C$. Only finitely many coefficients occur in the constructed polynomial differentials and augmentation. They therefore lie in some finite extension k'/k . The resulting augmented complex over $k'[x_1, \dots, x_d]$ becomes exact after extension to the algebraic closure; faithful flatness implies that it is already exact over $k'[x_1, \dots, x_d]$.

Choose a k -basis of k' . Restriction of scalars makes each free $k'[x]$ -module a free S -module of $[k' : k]$ times the rank, and identifies the underlying S -module of $M \otimes_k k'$ with $M^{\oplus [k' : k]}$. Thus we obtain a resolution of $M^{\oplus d[k' : k]}$. Each marked column over $k'[x]$ gives $[k' : k]$ columns over S , all still in $\mathfrak{m}^N$; entries in $\mathfrak{m}$ also remain in $\mathfrak{m}$. This proves the required rank, depth, and minimality assertions with $a = d[k' : k]$. $\square$

Remark 4.4. The use of Quillen–Suslin above is exactly the freeness of a finite projective module over a polynomial ring over a field. A recent self-contained route through Quillen’s and Suslin’s proofs is given by Dicks [5, Theorem 6].

5. PERIODIC DEGREE BOXES AND THE LOCAL THEOREM

Fix the polynomial resolution supplied by Proposition 4.3. Although it is exact, its entries need not be homogeneous. We introduce auxiliary variables u_i and replace x_i by $x_i u_i$. Assigning opposite degrees to x_i and u_i makes every substituted monomial homogeneous of degree zero. Restricting the auxiliary monomials to a large box then produces a finite graded free complex over S .

We use two boundary conditions for this box. The periodic box identifies opposite faces; it gives an exact complex and an exact length calculation, but does not preserve the grading. The open box discards terms leaving the box; it is graded, but can acquire homology. The proof compares their presentations near the boundary and controls the additional free modules needed to resolve the graded module. Both errors have size $O(L^{d-1})$, whereas the box contains L^d copies of every original basis vector.

Throughout this section, L alone tends to infinity. Constants in $O(L^{d-1})$ may depend on the fixed resolution and on N , as well as on the auxiliary integers chosen from them. They will disappear when we divide by L^d ; no uniformity over varying resolutions is needed in this construction.

Example (an open box in one variable). Let $S = k[x]$, let $M = S/(x^m)$ with $m \geq 1$, and take $L > m$. The original resolution has the single differential $S \xrightarrow{x^m} S$. Its open box has bases $e_0, \dots, e_{L-1}$ in degree one and $f_0, \dots, f_{L-1}$ in degree zero, with differential

$$d(e_\alpha) = \begin{cases} x^m f_{\alpha+m}, & \alpha + m < L, \\ 0, & \alpha + m \geq L. \end{cases}$$

Giving both e_α and f_α internal degree $-\alpha$ makes this map homogeneous of degree zero. The last m source vectors have zero image, while the first m target vectors receive no image. Every other target is hit by multiplication by x^m . Since that multiplication is injective on S , forgetting internal shifts gives

$$H_1(\mathcal{O}_L) \cong S^m, \quad H_0(\mathcal{O}_L) \cong S^m \oplus (S/(x^m))^{L-m}.$$

In particular, the open complex does not have finite-length homology. Taking the depth threshold and annihilator exponent both equal to m , the finite-length module obtained by imposing (x^m) is

$$H_0(\mathcal{O}_L)/x^m H_0(\mathcal{O}_L) \cong (S/(x^m))^L,$$

of length mL . The free homology comes from m cells at each of the two relevant boundaries, a number independent of L . This is why the general argument controls a correction to the open complex, rather than applying a finite-length estimate to its homology directly.

Proposition 5.1 (marked periodic boxes). *Let P be a polynomial free resolution, in degrees $0, \dots, d$, of a nonzero finite-length S -module M supported at $\mathfrak{m}$. Assume that P is minimal at $\mathfrak{m}$. In chosen bases, mark some positive-degree source columns N -deep, where $N \geq 1$, and let B be the total number of unmarked positive-degree source columns. There are finite-length graded S -modules M_L and choices of bases in their minimal graded resolutions such that, as $L \rightarrow \infty$,*

$$\ell_k(M_L) = L^d \ell_k(M) + O(L^{d-1}), \quad (5.1)$$

$$\beta_i^S(M_L) = L^d \text{rank}_S P_i + O(L^{d-1}), \quad (5.2)$$

$$B_N(M_L) \leq L^d B + O(L^{d-1}). \quad (5.3)$$

Here $B_N(M_L)$ is the total number of columns that are not N -deep in those chosen bases.

Proof. Step 1: construct the periodic and open boxes. Put $V_u = k[u_1, \dots, u_d]$, $R = S \otimes_k V_u$, and write $(x, u) = (x_1, \dots, x_d, u_1, \dots, u_d)$ and $(u) = (u_1, \dots, u_d)$. Consider

$$\phi : S \longrightarrow R, \quad x_i \longmapsto x_i u_i.$$

This map is flat. In one variable, $k[x_i, u_i]$ is torsion-free over $k[x_i]$ for this action: a nonzero polynomial in x_i maps to a nonzero polynomial in $x_i u_i$, which cannot annihilate a nonzero element of the domain $k[x_i, u_i]$. Torsion-free modules over a PID are flat. Taking the tensor product over k of these one-variable maps gives flatness of ϕ . Therefore

$$P^\Delta = R \otimes_{S, \phi} P \quad \text{resolves} \quad W = R \otimes_{S, \phi} M.$$

The matrices in P^Δ are obtained by the indicated substitution; its terms are free over R and hence flat over V_u .

For $L \geq 1$, let

$$T_L = V_u / (u_1^L, \dots, u_d^L), \quad T_L^{\text{per}} = V_u / (u_1^L - 1, \dots, u_d^L - 1).$$

Both have k -basis u^α with $0 \leq \alpha_i < L$, and both have dimension L^d . In T_L^{per} every u_i is a unit. Hence substitution by $x_i u_i$ is an automorphism of $S \otimes_k T_L^{\text{per}}$, with inverse substitution by $x_i u_i^{-1}$. The complex $P^\Delta \otimes_{V_u} T_L^{\text{per}}$ is consequently an automorphic image of the exact complex $P \otimes_k T_L^{\text{per}}$. As an S -complex it is free and resolves a module of length $L^d \ell_k(M)$. This argument uses only the units u_i ; it remains valid when the characteristic divides L and the periodic coefficient ring is not reduced.

The open complex is

$$\mathcal{O}_L = P^\Delta \otimes_{V_u} T_L. \quad (5.4)$$

It is finite free over S . To describe its grading and columns, give R the $\mathbb{Z}^d$ -grading $\deg x_i = e_i$, $\deg u_i = -e_i$, and give every original frame vector degree zero. A matrix term $c_\gamma x^\gamma$ of P acts on a box basis vector by

$$u^\alpha e_j \longmapsto c_\gamma x^\gamma u^{\alpha+\gamma} e_h,$$

unless some coordinate of $\alpha + \gamma$ reaches L , in which case this term is zero in the open box. The source and target have the same total multidegree, because the degrees of x^γ and u^γ cancel. Thus $\mathcal{O}_L$ is graded and has $L^d \text{rank } P_i$ homogeneous basis vectors in degree i .

Minimality of P at $\mathfrak{m}$ says that every entry of its positive differentials lies in $\mathfrak{m}$. Substitution and deletion of terms leaving the box do not lower the x -adic order of a column. Hence all columns of $\mathcal{O}_L$ are 1-deep, and its L^d copies of every marked column are N -deep.

The open complex need not resolve a finite-length module. Choose $C \geq N$ with $\mathfrak{m}^C M = 0$, and impose the same annihilator on its zeroth homology by setting

$$U = W/\mathfrak{m}^C W, \quad M_L = U \otimes_{V_u} T_L = H_0(\mathcal{O}_L)/\mathfrak{m}^C H_0(\mathcal{O}_L). \quad (5.5)$$

The equality follows from right exactness of tensor product, since $H_0(\mathcal{O}_L) = W \otimes_{V_u} T_L$. The module M_L is finite over $S/\mathfrak{m}^C$, and hence has finite length. It retains the $\mathbb{Z}^d$ -grading, which we subsequently coarsen by total degree to the usual grading over S .

Most of the annihilator relations in (5.5) are already present in the open complex. For $|\gamma| = C$ and each original generator e_j of P_0 , exactness of P gives a polynomial column $h_{\gamma,j}$ such that $d_1 h_{\gamma,j} = x^\gamma e_j$. Substitution yields

$$d_1(xu)h_{\gamma,j}(xu) = x^\gamma u^\gamma e_j.$$

If every coordinate of α is at least C , then $\alpha - \gamma$ has nonnegative coordinates. Multiplying the last identity by $u^{\alpha-\gamma}$ and passing to the open box shows that $x^\gamma u^\alpha e_j$ is already a boundary. Since the monomials of degree C generate $\mathfrak{m}^C$, no new annihilator relations are needed at such a cell. New relations are required only at cells with some coordinate less than C , whose number is

$$L^d - (L - C)^d = O(L^{d-1}) \quad (L \geq C). \quad (5.6)$$

All the old relations lie in $\mathfrak{m}$, and all the added relations lie in $\mathfrak{m}^C \subseteq \mathfrak{m}$. Reducing this presentation modulo $\mathfrak{m}$ therefore gives a k -space of dimension $L^d \text{rank } P_0$. In particular, $M_L \neq 0$ and it has exactly that many minimal generators.

Step 2: compare lengths by deleting a fixed boundary. Let E bound each coordinate exponent of every monomial occurring in d_1 , set $b = \max\{C, E\}$, and take $L > b$. The periodic module is killed by $\mathfrak{m}^C$: before twisting it is $M \otimes_k T_L^{\text{per}}$, and the twisting automorphism preserves the extended ideal $\mathfrak{m}^C$. Thus both the periodic module and M_L have presentations with the same coefficient module

$$A_L = (S/\mathfrak{m}^C)^{L^d \text{rank } P_0}.$$

Let A_L^∂ be the span of target cells whose index has some coordinate less than b . Compare the two presentations after quotienting A_L by this boundary submodule.

For a source cell α , a matrix monomial x^γ either has target $\alpha + \gamma$ inside the box or leaves the box. In the former case the open and periodic matrices agree. In the latter case the open matrix discards the term. In any coordinate i where the periodic term wraps, its target coordinate is

$$\alpha_i + \gamma_i - L < \gamma_i \leq E \leq b.$$

There is at most one wrap in a coordinate because $\gamma_i < L$. The wrapped target is therefore in A_L^∂ and is zero in the quotient. It follows that the two relation matrices induce the same presentation over A_L/A_L^∂ . Denote the common cokernel by I_L .

Both modules surject onto I_L . In each case the kernel is an image of A_L^∂ , so its length is at most

$$\ell_k A_L^\partial = \text{rank}(P_0)(L^d - (L - b)^d) \ell_k(S/\mathfrak{m}^C).$$

Since the right side is $O(L^{d-1})$, the two module lengths differ by $O(L^{d-1})$. The periodic length is exactly $L^d \ell_k(M)$, which proves (5.1). This comparison uses presentations only; it makes no assertion about the positive homology of the open complex.

Step 3: bound the free rank needed to repair the open complex. We now estimate the total Betti number of $\text{Cone}(\mathcal{O}_L \rightarrow M_L)$. A small value of this invariant will allow us to apply Lemma 2.4. There are two sources of error: imposing $\mathfrak{m}^C$ on W , and replacing a derived tensor product by its zeroth homology. We treat them separately.

Put

$$K = \mathfrak{m}^C W, \quad v = u_1 \cdots u_d, \quad Y_L = U \otimes_{V_u}^{\mathbf{L}} T_L.$$

For $|\gamma| = C$, the original annihilator relation implies that $x^\gamma u^\gamma$ annihilates W . Moreover u^γ divides v^C , since each $\gamma_i \leq C$. It follows that $v^C x^\gamma W = 0$ for each generator x^γ of $\mathfrak{m}^C$, and hence $v^C K = 0$.

The complex P^Δ is a V_u -flat resolution of W , so $\mathcal{O}_L$ represents $W \otimes_{V_u}^{\mathbf{L}} T_L$. Apply derived tensor with T_L to $0 \rightarrow K \rightarrow W \rightarrow U \rightarrow 0$. The resulting triangle identifies the first comparison cone as

$$\text{Cone}(\mathcal{O}_L \rightarrow Y_L) \simeq (K \otimes_{V_u}^{\mathbf{L}} T_L)[1]. \quad (5.7)$$

The shift does not change total Betti number.

To estimate that number, take the derived residue over S . Put

$$D_L = R/(x_1, \dots, x_d, u_1^L, \dots, u_d^L) \cong T_L.$$

Associativity of derived tensor products gives

$$k \otimes_S^{\mathbf{L}} (K \otimes_{V_u}^{\mathbf{L}} T_L) \simeq K \otimes_R^{\mathbf{L}} D_L.$$

One can see the identity by taking a free R -resolution of K : tensoring it first over V_u with T_L gives a free S -complex, and then reducing over S by $\mathfrak{m}$ gives its tensor with D_L . Thus the dimensions of the homology on the right are exactly the S -Betti numbers on the left.

The ring D_L is Artinian local with residue field k . Every element of R outside (x, u) maps to a unit of D_L , since its constant term is nonzero. We may consequently localize at (x, u) before computing the last derived tensor product. The fixed finite module $K_{(x,u)}$ has a finite free resolution over the regular local ring $R_{(x,u)}$. Let B_K be its total free rank. The integer B_K is fixed before L is varied.

Multiplication by v^C on this resolution lifts the zero map on $K_{(x,u)}$. The comparison theorem for projective resolutions therefore makes it chain nullhomotopic. Tensoring with D_L preserves that homotopy. Lemma 2.5, applied to this reduced free complex, gives

$$\begin{aligned} \beta_S^{\text{tot}}(K \otimes_{V_u}^{\mathbf{L}} T_L) &\leq B_K \ell_k V_u / (v^C, u_1^L, \dots, u_d^L) \\ &= B_K (L^d - (L - C)_+^d). \end{aligned} \quad (5.8)$$

Here $z_+ = \max\{z, 0\}$. The last equality is a monomial count: among the L^d monomials u^α in T_L , exactly $(L - C)_+^d$ are divisible by v^C . In particular, the bound is $O(L^{d-1})$ for the fixed C . This computation takes place over R and its Artinian quotient. It does not require K to be finite over S or supported at $\mathfrak{m}$ as an S -module.

Next consider the positive homology of Y_L . The module U is finite over V_u , because it is finite over $R/\mathfrak{m}^C$ and that ring is finite free over V_u . After inverting v , the substitution by $x_i u_i$ is an automorphism. It preserves $\mathfrak{m}^C$, so $\mathfrak{m}^C W_v = 0$ and

$$U_v = W_v \cong M \otimes_k V_{u,v} \cong V_{u,v}^{\ell_k(M)}$$

as $V_{u,v}$ -modules.

We use this freeness to isolate the part supported where $v = 0$. Take a basis of U_v and multiply its elements by sufficiently large powers of v so that they come from elements of U . They define a map $V_u^{\ell_k(M)} \rightarrow U$ which becomes an isomorphism after inverting v . Its kernel is v -torsion in a free module over the domain V_u , and is therefore zero. Write its image as Λ and its cokernel as Q . We have

$$0 \rightarrow \Lambda \rightarrow U \rightarrow Q \rightarrow 0, \quad \Lambda \cong V_u^{\ell_k(M)}.$$

Since Q is finite and $Q_v = 0$, there is a fixed H such that $v^H Q = 0$. This lattice is only a V_u -module lattice; it need not be stable under the S -action, which is not used in this step.

The long exact Tor sequence, together with the freeness of Λ , gives

$$\text{Tor}_1^{V_u}(U, T_L) \hookrightarrow \text{Tor}_1^{V_u}(Q, T_L), \quad \text{Tor}_j^{V_u}(U, T_L) \cong \text{Tor}_j^{V_u}(Q, T_L) \quad (j > 1).$$

Since T_L is supported at (u) , these Tor groups can be computed after localization there. Choose a fixed finite free resolution of $Q_{(u)}$ over the regular local ring $(V_u)_{(u)}$, and let B_U be its total rank. Multiplication by v^H is nullhomotopic on this resolution because it is zero on $Q_{(u)}$. Applying Lemma 2.5 after reduction to T_L and using the same monomial count gives

$$\sum_{j>0} \ell_k \operatorname{Tor}_j^{V_u}(U, T_L) \leq B_U(L^d - (L - H)_+^d) = O(L^{d-1}). \quad (5.9)$$

These Tor modules carry the S -action induced from U . Every element of $\mathfrak{m}^C$ acts by the zero map on U , so functoriality of Tor shows that it also acts by zero on these groups. They therefore have finite S -length, equal to their k -dimension. The natural map $Y_L \rightarrow H_0(Y_L) = M_L$ induces an isomorphism in degree zero. Its cone has homology

$$H_{j+1}(\operatorname{Cone}(Y_L \rightarrow M_L)) \cong \operatorname{Tor}_j^{V_u}(U, T_L) \quad (j > 0),$$

and no other homology. Lemma 2.2 consequently gives the explicit estimate

$$\beta_S^{\operatorname{tot}} \operatorname{Cone}(Y_L \rightarrow M_L) \leq 2^d \sum_{j>0} \ell_k \operatorname{Tor}_j^{V_u}(U, T_L) = O(L^{d-1}).$$

Finally, the two composable maps $\mathcal{O}_L \rightarrow Y_L \rightarrow M_L$ give a triangle relating their two individual cones to the cone of the composite. Subadditivity of total Betti number in this triangle yields

$$\begin{aligned} \beta_S^{\operatorname{tot}} \operatorname{Cone}(\mathcal{O}_L \rightarrow M_L) &\leq \beta_S^{\operatorname{tot}} \operatorname{Cone}(\mathcal{O}_L \rightarrow Y_L) + \beta_S^{\operatorname{tot}} \operatorname{Cone}(Y_L \rightarrow M_L) \\ &= O(L^{d-1}), \end{aligned}$$

by (5.7), (5.8), and (5.9). With $C_L^{\operatorname{box}} = \operatorname{Cone}(\mathcal{O}_L \rightarrow M_L)$, we have proved

$$\beta_S^{\operatorname{tot}}(C_L^{\operatorname{box}}) = O(L^{d-1}). \quad (5.10)$$

Step 4: minimize while retaining the marked columns. We now check the hypotheses needed to convert the cone estimate into a statement about actual resolutions. The complex $\mathcal{O}_L$ is bounded graded free in nonnegative degrees, and all its differential entries belong to $\mathfrak{m}$. The module M_L is finite and graded over the regular ring S , so it has finite projective dimension. The natural map $\mathcal{O}_L \rightarrow M_L$ is homogeneous of degree zero and is surjective on zeroth homology. Thus the graded form of Lemma 2.4 applies.

Let E_L be a minimal graded free model of C_L^{box} . It starts in degree one because the zeroth-homology map is surjective. The lemma supplies a graded free resolution Z_L of M_L in a degreewise split sequence

$$0 \longrightarrow \mathcal{O}_L \longrightarrow Z_L \longrightarrow E_L \longrightarrow 0, \quad \sum_i \operatorname{rank}_S E_{L,i} = O(L^{d-1}). \quad (5.11)$$

The inclusion is an inclusion of complexes, not merely a rank comparison: each column inherited from $\mathcal{O}_L$ has its original entries and zero components in the added target summands. This is why the selected column depths can be carried into the minimization argument.

Write $\eta_L = \beta_S^{\operatorname{tot}}(C_L^{\operatorname{box}}) = O(L^{d-1})$. The rank and column bounds of Lemma 2.4 are

$$\left| \beta_i^S(M_L) - L^d \operatorname{rank}_S P_i \right| \leq \eta_L, \quad B_N(M_L) \leq B_N(\mathcal{O}_L) + \eta_L \leq L^d B + \eta_L.$$

The last inequality uses the L^d retained copies of every marked column; some unmarked copies may also become deep after truncation, which only improves the bound. The lemma produces homogeneous bases in the minimal graded resolution, so these are precisely the basis choices required in the proposition. This proves (5.2) and (5.3). $\square$

Remark 5.2. The correction E_L has small total free rank, but its grading shifts can grow with L . It must therefore be retained until after Theorem 3.6 has been applied. The following calculation makes the distinction explicit. For a graded free complex with terms $C_i = \bigoplus_j S(-j)^{r_{i,j}(C)}$, put

$$P_C(z) = \sum_{i,j} (-1)^i r_{i,j}(C) z^j.$$

Here $r_{i,j}(C)$ records the chosen free terms, whether or not the complex is minimal. Every homological degree of $\mathcal{O}_L$ uses the same list of box shifts, each repeated rank P_i times. Since the resolution P resolves a torsion S -module, $\sum_i (-1)^i \text{rank } P_i = 0$. Consequently

$$P_{\mathcal{O}_L}(z) = \left(\sum_{0 \leq \alpha_j < L} z^{-|\alpha|} \right) \left(\sum_i (-1)^i \text{rank } P_i \right) = 0.$$

The split sequence (5.11) gives $P_{Z_L} = P_{E_L}$. Since Z_L resolves M_L , its Hilbert numerator is $(1-z)^d \text{Hilb}_{M_L}(z)$, and hence

$$P_{E_L}^{(d)}(1) = (-1)^d d! \ell_k(M_L).$$

Thus a correction of total term rank $O(L^{d-1})$ carries a d th moment of order L^d . Rank estimates alone would not justify discarding that moment. The sparse-column theorem handles the correction through its exceptional columns, with no bound on their shifts.

Proof of Theorem 1.3. The case $M = 0$ is immediate. For $M \neq 0$, complete the local ring and the chosen resolution. Completion is faithfully flat, preserves length and minimality, and leaves the Betti numbers unchanged. It also preserves each specified column order: for a finite free target, membership in $\mathfrak{m}^N$ can be checked after completion, since $\mathfrak{m}^N \hat{A} \cap A = \mathfrak{m}^N$. The Cohen structure theorem identifies the completion of the equicharacteristic regular local ring with a power-series ring over a coefficient field. We may therefore assume $A = k[[x_1, \dots, x_d]]$.

Mark all the N -deep columns in the chosen minimal resolution. Proposition 4.3 gives a polynomial resolution of $M^{\oplus a}$ for some positive integer a , with ranks $a\beta_i(M)$ and at most aB unmarked columns. Fix this polynomial resolution and all the auxiliary choices made from it. Proposition 5.1 then produces graded modules M_L with

$$\begin{aligned} \ell_k(M_L) &= aL^d \ell_A(M) + O(L^{d-1}), \\ \beta_r^S(M_L) &= aL^d \beta_r^A(M) + O(L^{d-1}), \\ B_N(M_L) &\leq aL^d B + O(L^{d-1}). \end{aligned}$$

Apply Theorem 3.6 to each M_L in the bases provided by the box proposition. Because the exceptional-column term is subtracted, replacing $B_N(M_L)$ by its upper bound gives

$$\begin{aligned} aL^d \ell_A(M) + O(L^{d-1}) &\geq \frac{N^d}{\binom{d}{r}} (aL^d \beta_r^A(M) + O(L^{d-1})) \\ &\quad - C_d N^d (aL^d B + O(L^{d-1})). \end{aligned}$$

Divide by aL^d and let $L \rightarrow \infty$, keeping M , N , a , and the polynomial resolution fixed. Every error becomes $O(L^{-1})$ and vanishes. We obtain

$$\ell_A(M) \geq \frac{N^d}{\binom{d}{r}} \beta_r^A(M) - C_d N^d B,$$

which is (1.3). The surviving constant is exactly the constant of the graded theorem; it does not depend on any globalization or box choices. This order of limits is essential: the box limit is

completed for each fixed module before the local inequality is applied to a moving family in later sections.

For completeness, the total-rank inequality follows from a weighted average, rather than an unweighted sum. Multiply the row- r inequality by $\binom{d}{r}/2^d$ and sum over $r = 0, \dots, d$. These weights sum to one, so the length and error terms remain unchanged, while

$$\sum_{r=0}^d \frac{\binom{d}{r}}{2^d} \frac{N^d \beta_r^A(M)}{\binom{d}{r}} = \left(\frac{N}{2}\right)^d \sum_{r=0}^d \beta_r^A(M).$$

This proves (1.4). $\square$

We now have the local sparse-column inequality with constants independent of the ranks and grading shifts of the resolution. The next section constructs a moving family to which this local inequality can be applied. Its task is to turn a comparison controlled by one scalar into resolutions in which the proportion of exceptional columns tends to zero.

6. COORDINATE-POWER RESTRICTION AND SCALAR-CLASS MASS

Let $A = k[[x_1, \dots, x_d]]$, with k an arbitrary field and $d \geq 1$. For an integer $t \geq 1$, define

$$\sigma_t : A \longrightarrow A, \quad \sigma_t(x_i) = x_i^t, \quad \sigma_t|_k = \text{id}_k,$$

and denote restriction of scalars by σ_{t*} . Thus x_i acts on $\sigma_{t*}V$ as multiplication by x_i^t on V , whereas the coefficient field acts unchanged. The distinction between free rank and finite length under this operation will be essential below: the operation enlarges a free module by a factor of t^d , but it does not change the length of a finite-length module. We first establish these facts and then estimate what happens to a comparison cone and to differential depth.

Lemma 6.1 (coordinate restriction). *The map σ_t is finite free of rank t^d . Restriction is exact, turns a free module of rank b into one of rank bt^d , and for every finite-length module V ,*

$$\ell_A(\sigma_{t*}V) = \ell_A(V). \quad (6.1)$$

If C is a bounded complex of finite free modules and $J_t = (x_1^t, \dots, x_d^t)$, then

$$\beta_A^{\text{tot}}(\sigma_{t*}C) = \dim_k H_{\text{tot}}(C \otimes_A A/J_t). \quad (6.2)$$

Proof. Grouping monomials by their exponent residues gives

$$A = \bigoplus_{0 \leq \alpha_i < t} x^\alpha \sigma_t(A).$$

This proves finite freeness and the rank assertion. Restriction of scalars does not change the underlying abelian groups, so it is exact. A composition series proves (6.1): restriction preserves each short exact sequence in the series, and each factor remains the simple module k because σ_t fixes k and maps $\mathfrak{m}$ into $\mathfrak{m}$. Finally, quotienting $\sigma_{t*}C$ by its maximal ideal means quotienting the original complex C by the action of $x_1^t, \dots, x_d^t$. Hence

$$k \otimes_A \sigma_{t*}C \cong C \otimes_A A/J_t$$

as complexes of k -vector spaces. The restricted complex is free, so the left side computes $k \otimes_A^{\mathbf{L}} \sigma_{t*}C$. Taking its homology in each degree and then summing the dimensions proves (6.2). No minimality assumption on C is needed for this identity. $\square$

Lemma 6.2 (restricted scalar cone). *Let C be a bounded complex of finite free A -modules of total term rank at most B . Suppose multiplication by $0 \neq a \in A$ is nullhomotopic on C , and put $h = \text{ord}_{\mathfrak{m}}(a)$. Then*

$$\beta_A^{\text{tot}}(\sigma_{t*}C) \leq B \ell_A(A/(a, J_t)) \leq Bh \binom{dt-1}{d-1}. \quad (6.3)$$

If a differential has entries in $\mathfrak{m}^N$, its restriction has entries in $\mathfrak{m}^n$ whenever

$$N \geq tn + d(t-1). \quad (6.4)$$

Proof. Reduce the given nullhomotopy modulo J_t and apply Lemma 2.5 over the Artinian ring $Q = A/J_t$ to the free Q -complex $\overline{C} = C \otimes_A Q$. The hypothesis survives reduction because the identity $dh + hd = a \text{id}_C$ remains an identity modulo J_t . The lemma gives

$$\sum_i \ell_Q H_i(\overline{C}) \leq B \ell_Q(Q/aQ).$$

Both lengths here are also dimensions over the unchanged coefficient field k . Equation (6.2) now proves the first inequality in (6.3). This application takes place over Q with the original scalar a ; it does not require a to lie in the image of σ_t .

Set $L = d(t-1) + 1$. Then $\mathfrak{m}^L \subseteq J_t$. Because $\text{gr}_{\mathfrak{m}}(A)$ is a domain, the initial ideal of the principal ideal (a) is generated by the initial form of a , of degree h . Indeed, $\text{ord}(af) = h + \text{ord}(f)$ and $\text{in}(af) = \text{in}(a)\text{in}(f)$ for every nonzero $f \in A$. For $h > 0$ the degree- r Hilbert function of this hypersurface is

$$\binom{r+d-1}{d-1} - \binom{r-h+d-1}{d-1},$$

where the second term is zero for $r < h$. Summing from $r = 0$ to $L-1$ cancels all but at most h terms of the first sequence. Every remaining term is at most $\binom{L+d-2}{d-1}$, and therefore

$$\ell A/(a, J_t) \leq \ell A/(a, \mathfrak{m}^L) \leq h \binom{L+d-2}{d-1} = h \binom{dt-1}{d-1}.$$

When a is a unit both sides are zero.

For (6.4), a monomial x^γ of total degree at least $tn + d(t-1)$ satisfies

$$\sum_i \left\lfloor \frac{\gamma_i}{t} \right\rfloor \geq n.$$

Thus $\mathfrak{m}^N \subseteq \sigma_t(\mathfrak{m}^n)A$. Multiplication by a basis monomial x^α , $0 \leq \alpha_i < t$, cannot decrease the sum of the exponent quotients. More explicitly, write $\gamma_i + \alpha_i = t\delta_i + \epsilon_i$, with $0 \leq \epsilon_i < t$. In the restriction basis the resulting term is $\sigma_t(x^\delta)x^\epsilon$, and $|\delta| \geq n$. Applying this to every monomial of every matrix entry proves that each coefficient of the restricted matrix lies in the new maximal-ideal power $\mathfrak{m}^n$. $\square$

The next theorem converts a comparison with a deep free complex into a lower bound for the length of zeroth homology. There are two errors to absorb: the scalar-controlled comparison cone and the positive homology of the target. Coordinate restriction makes their total Betti numbers negligible relative to the inherited free ranks.

Theorem 6.3 (scalar-class mass). *Let q range through a sequence tending to infinity. Suppose:*

- (1) G_q is a free complex in degrees $0, \dots, d$, with fixed term ranks b_i and $b_0 > 0$, and every differential entry lies in $\mathfrak{m}^{N_q}$, where $N_q/q \rightarrow \rho$ for a fixed $\rho > 0$;
- (2) Y_q is a family of free complexes with uniformly bounded total term rank and amplitude, with finite-length homology and $H_j(Y_q) = 0$ for $j < 0$;
- (3) there is a chain map $G_q \rightarrow Y_q$ whose cone is nullhomotopic under a nonzero scalar a_q , where $h_q = \text{ord}_{\mathfrak{m}}(a_q) = o(q)$;
- (4) for $M_q = H_0(Y_q)$,

$$s_q = \sum_{j>0} \ell_A H_j(Y_q) = o(q^d).$$

Then, for $0 \leq i \leq d$,

$$\liminf_q \frac{\ell_A(M_q)}{q^d} \geq \rho^d \frac{b_i}{\binom{d}{i}}. \quad (6.5)$$

Moreover,

$$\liminf_q \frac{\ell_A(M_q)}{q^d} \geq \frac{\rho^d}{2^d} \sum_{i=0}^d b_i. \quad (6.6)$$

Proof. Choosing one scale for both errors. We first choose a restriction scale that is small relative to q but large relative to both error terms. Put

$$u_q = \max\{1, h_q, (s_q + 1)^{1/d}\}, \quad t_q = \lceil \sqrt{qu_q} \rceil.$$

The hypotheses give $u_q/q \rightarrow 0$. The definition of t_q gives

$$\frac{t_q}{q} \leq \sqrt{\frac{u_q}{q}} + \frac{1}{q}, \quad \frac{u_q}{t_q} \leq \sqrt{\frac{u_q}{q}}.$$

Both ratios therefore tend to zero. Since $h_q/t_q \leq u_q/t_q$ and $s_q/t_q^d \leq (u_q/t_q)^d$, we obtain

$$t_q \rightarrow \infty, \quad t_q = o(q), \quad h_q = o(t_q), \quad s_q = o(t_q^d). \quad (6.7)$$

Separating the two comparison errors. Let $C_q = \text{Cone}(G_q \rightarrow Y_q)$. We need to compare G_q with the module M_q , rather than with the whole complex Y_q . To construct the map to that module, first replace the degree-zero term of Y_q by $\ker(d_0)$ and omit the negative terms. This truncated complex maps to Y_q by inclusion and is quasi-isomorphic to it, since the negative homology vanishes. It also maps onto $\ker(d_0)/\text{im}(d_1) = M_q$ in degree zero. The two maps define the morphism $Y_q \rightarrow M_q$ in the derived category. As Y_q is bounded free, this morphism can be represented by a chain map from Y_q itself.

Let $Q_q = \text{Cone}(Y_q \rightarrow M_q)$ and $K_q = \text{Cone}(G_q \rightarrow M_q)$, using the composite chain map for the latter. The map on zeroth homology is the identity of M_q , so the long exact homology sequence of the first cone gives

$$H_i(Q_q) \cong H_{i-1}(Y_q) \quad (i \geq 2), \quad H_i(Q_q) = 0 \quad (i \leq 1).$$

Thus Q_q records exactly the positive homology discarded from Y_q , with each group shifted up by one degree.

There is a natural map $C_q \rightarrow K_q$: it applies $Y_q \rightarrow M_q$ on the target summand of the cone and the identity on its shifted G_q summand. In the cone of this map the two copies of G_q are joined by an identity map. Cancelling that contractible summand leaves Q_q . We therefore obtain the triangle

$$C_q \rightarrow K_q \rightarrow Q_q \rightarrow C_q[1].$$

This is the usual triangle for two composable maps, here keeping track of the two errors separately.

The total term ranks of C_q have a fixed bound B' , since those of both G_q and Y_q do. Lemma 6.2 bounds the total Betti number of $\sigma_{t_q*}C_q$ by $B'h_q\binom{dt_q-1}{d-1}$. On the other hand, exactness of restriction and preservation of finite length show that the sum of the homology lengths of $\sigma_{t_q*}Q_q$ is exactly s_q . Lemma 2.2 therefore bounds its total Betti number by $2^d s_q$. Applying derived tensor with k to the displayed triangle and using its long exact homology sequence gives

$$\eta_q := \beta_A^{\text{tot}}(\sigma_{t_q*}K_q) \leq B'h_q\binom{dt_q-1}{d-1} + 2^d s_q = o(t_q^d). \quad (6.8)$$

For the last equality, divide the first summand by t_q^d and use $\binom{dt_q-1}{d-1} = O(t_q^{d-1})$ together with $h_q/t_q \rightarrow 0$; the second summand is handled by $s_q/t_q^d \rightarrow 0$.

Recovering a resolution with few exceptional columns. The regularity of A makes M_q perfect. We will apply Lemma 2.4 to the restricted morphism $\sigma_{t_q*}G_q \rightarrow \sigma_{t_q*}M_q$, whose comparison cone has total Betti number η_q by (6.8). First record the depth of its inherited columns.

Set

$$n_q = \max \left\{ 1, \left\lfloor \frac{N_q - d(t_q - 1)}{t_q} \right\rfloor \right\}. \quad (6.9)$$

For large q the maximum is unnecessary, and

$$\frac{n_q t_q}{q} \longrightarrow \rho. \quad (6.10)$$

Indeed, $N_q/t_q \rightarrow \infty$, and rounding down changes $n_q t_q$ by less than t_q . The further subtraction $d(t_q - 1)$ is also $o(q)$. Hence $n_q t_q = N_q + o(q)$. For large q , equation (6.4) shows that every source column of $\sigma_{t_q*}G_q$ is n_q -deep. In particular its differentials lie in $\mathfrak{m}$, as required by Lemma 2.4. The lemma gives a minimal resolution P_q of $\sigma_{t_q*}M_q$ with

$$|\beta_i(P_q) - t_q^d b_i| \leq \eta_q, \quad B_{n_q}(P_q) \leq \eta_q.$$

Here the inherited free summand in degree i has rank $t_q^d b_i$ by Lemma 6.1. The comparison lemma adds a free model of the cone, of total rank η_q , before cancelling unit pivots. Its column estimate says that this operation can introduce at most η_q exceptional columns; the depths of the surviving inherited columns are retained. This is why controlling the total Betti number of the cone, rather than just its homology length, was the needed intermediate estimate. In particular,

$$\beta_i(P_q) = t_q^d b_i + o(t_q^d), \quad B_{n_q}(P_q) = o(t_q^d). \quad (6.11)$$

For large q it is nonzero, since $\beta_0(P_q) \geq t_q^d b_0 - \eta_q > 0$.

Applying the uniform local inequality. Apply Theorem 1.3 to P_q . By (6.1), its module has length $\ell_A(M_q)$. Writing C_d for the constant in that theorem gives the explicit inequality

$$\frac{\ell_A(M_q)}{q^d} \geq \left(\frac{n_q t_q}{q} \right)^d \left(\frac{\beta_i(P_q)}{t_q^d \binom{d}{i}} - C_d \frac{B_{n_q}(P_q)}{t_q^d} \right).$$

The leading term combines the inherited rank $t_q^d b_i$ with the d th power of the converted depth n_q . After division by q^d , its scale is

$$\frac{n_q^d t_q^d}{q^d} \longrightarrow \rho^d,$$

and the exceptional term tends to zero because

$$\frac{n_q^d B_{n_q}(P_q)}{q^d} = \left(\frac{n_q t_q}{q} \right)^d \frac{B_{n_q}(P_q)}{t_q^d} \longrightarrow 0.$$

This proves (6.5); the total form of Theorem 1.3 proves (6.6). All choices here are functions of the single index q . The local inequality has already been proved with a constant independent of the module, its ranks, and n_q , so it applies to these moving resolutions without any further interchange of limits. $\square$

Remark 6.4. Coordinate restriction preserves finite length but multiplies each free rank by t_q^d . It changes differential depth from approximately ρq to approximately $\rho q/t_q$. Consequently the product of free rank and the d th power of depth remains on the q^d scale. The choice of t_q makes the exceptional columns a vanishing fraction of the inherited columns.

The scalar in Theorem 6.3 acts by an actual nullhomotopy on a bounded free cone. Merely annihilating the cone's homology would not give (6.3). Coordinate-power restriction is separate from Frobenius and works over every field.

7. STRICT MULTIPLICITY LATTICES

Let $(D, \mathfrak{n}, k)$ be a complete equicharacteristic local ring of dimension $d \geq 1$, with an infinite coefficient field k . No domain, reducedness, equidimensionality, or Cohen–Macaulay hypothesis is made. The construction uses classical graded Noether normalization and the associativity formula for multiplicity. We choose a homogeneous basis of $\text{gr}_{\mathfrak{n}}(D)$ over the fraction field of a linear normalization, and then lift its elements to D . Independence of their initial forms prevents cancellation after lifting. The resulting free submodule has rank $e(D)$ and an explicitly controlled induced $\mathfrak{n}$ -adic filtration.

Proposition 7.1 (strict Noether lattice). *There is a finite injective local map*

$$A = k[[y_1, \dots, y_d]] \longrightarrow D$$

and elements $b_1, \dots, b_e \in D$, with integers $a_j \geq 0$, such that $e = e(D)$ and

$$L := \bigoplus_{j=1}^e A b_j \subseteq D, \quad L \cap \mathfrak{n}^u D = \bigoplus_{j=1}^e \mathfrak{m}_A^{\max\{u-a_j, 0\}} b_j \quad (u \geq 0). \quad (7.1)$$

Moreover D/L is a finite torsion A -module. Consequently there is $0 \neq c \in A$ with $cD \subseteq L$. If $f : L \hookrightarrow D$ is inclusion and $g : D \rightarrow L$ is multiplication by c , then

$$fg = \text{id}_D, \quad gf = \text{id}_L. \quad (7.2)$$

Proof. A free submodule in the associated graded ring. Put $B = \text{gr}_{\mathfrak{n}}(D)$. Since k is infinite, choose a linear homogeneous Noether normalization

$$P = k[Y_1, \dots, Y_d] \hookrightarrow B.$$

The ring B is a finite graded P -module. Choose a finite homogeneous generating set and select elements b_j^* whose images form a basis after tensoring with the fraction field of P ; write $a_j = \deg b_j^*$. Any P -linear relation among these elements would remain a relation over the fraction field, so their P -span is $\bigoplus_j P(-a_j)$. After tensoring with the fraction field this span becomes all of B , and hence its cokernel is P -torsion.

A finite torsion P -module has dimension less than d . Its Hilbert function therefore has lower growth order and contributes nothing to the leading term used to compute multiplicity. Each shifted copy $P(-a_j)$ has multiplicity one: a fixed degree shift does not change the leading coefficient. Consequently the number of chosen elements is

$$\text{rank}_P B = e(B) = e(D).$$

Here the degree-one hypothesis on the variables of P is essential. Lift Y_i to $y_i \in \mathfrak{n}$ and b_j^* to $b_j \in \mathfrak{n}^{a_j}$.

Lifting the parameter ring and its free submodule. Finiteness of B over P means that, in all sufficiently large degrees, $B_{r+1} = \sum_i Y_i B_r$. After lifting to D and applying Nakayama's lemma this gives $\mathfrak{n}^{r+1} = Q \mathfrak{n}^r$ for large r , where $Q = (y_1, \dots, y_d)$. Thus Q is a reduction of $\mathfrak{n}$ and is $\mathfrak{n}$ -primary. The lifts y_i define a continuous map from the power series ring $A = k[[y_1, \dots, y_d]]$ to D . Since D is complete and D/QD is finite-dimensional over k , lifts of a basis of D/QD generate D as an A -module: successive expansion modulo powers of Q gives coefficients that converge in A . The map is therefore finite. Finiteness implies $\dim(A/\ker(A \rightarrow D)) = \dim D = d$. A nonzero ideal in the d -dimensional regular local domain A has quotient of smaller dimension, so this kernel is zero.

Take coefficients $f_j \in A$, not all zero, and put $z = \sum_j f_j b_j$. Let

$$w = \min_j \{\text{ord}_{\mathfrak{m}_A}(f_j) + a_j\}.$$

The element z belongs to $\mathfrak{n}^w D$, and its image in $\mathfrak{n}^w D / \mathfrak{n}^{w+1} D$ is

$$\sum_{\text{ord}_{\mathfrak{m}_A}(f_j) + a_j = w} \text{in}(f_j) b_j^*.$$

It is nonzero because the b_j^* are P -linearly independent. Thus $z \neq 0$ and $\text{ord}_{\mathfrak{n}}(z) = w$. This proves that the lifted elements are A -linearly independent. It also identifies membership in $\mathfrak{n}^u D$ exactly: $\sum_j f_j b_j \in \mathfrak{n}^u D$ if and only if $\text{ord}_{\mathfrak{m}_A}(f_j) + a_j \geq u$ for every j . These individual conditions are precisely $f_j \in \mathfrak{m}_A^{\max\{u-a_j, 0\}}$, giving the intersection formula (7.1). In particular, the formula controls the filtration induced from D on L , not just an abstract filtration on the free module L .

Clearing the torsion quotient. Since Q is a reduction, $e_Q(D) = e_{\mathfrak{n}}(D)$. For the finite A -module D , the associativity formula over the regular domain A [28, Lemma 43.15.3] gives

$$\text{rank}_A D = e_Q(D) = e(D) = \text{rank}_A L.$$

To spell out the first equality, the only prime of A with d -dimensional quotient is (0) , and its contribution to the associativity formula is $\dim_{\text{Frac}(A)}(D \otimes_A \text{Frac}(A))$, since $e(A) = 1$. Moreover $\mathfrak{m}_A D = QD$, and the common residue field makes lengths of finite-length D -modules the same over A and D . Thus the A -module multiplicity in that formula is exactly $e_Q(D)$.

The equal-rank inclusion $L \subseteq D$ becomes an isomorphism after tensoring with $\text{Frac}(A)$, so D/L is torsion. Choose a finite set of generators of D/L and a nonzero element of A annihilating each generator. Their product is a nonzero element c annihilating the whole quotient. Therefore $cD \subseteq L$, which defines the map g . Both composites in (7.2) are multiplication by c . This argument allows D itself to have A -torsion; in particular, c need not be a nonzerodivisor on D . $\square$

Let F be a minimal short free D -complex and write $\beta_i = \text{rank}_D F_i$. For $q = p^n$ in characteristic p , denote by $F_q = \mathcal{F}_D^n(F)$ the Frobenius pullback: in fixed bases its differential matrices are obtained by raising every entry to the q th power. We will need its homology to remain supported at the maximal ideal. For any prime $\mathfrak{p} \neq \mathfrak{n}$, the complex $F_{\mathfrak{p}}$ is bounded, free, and acyclic. Such a complex is split exact: starting at its lowest nonzero degree, split the surjection onto the bottom free module and continue upward. Frobenius extension preserves these splittings, so $(F_q)_{\mathfrak{p}}$ is also exact. Each $H_i(F_q)$ is therefore a finite module supported only at $\mathfrak{n}$, and hence has finite length.

Proposition 7.2 (lattice compression). *With the notation of Proposition 7.1, define an A -free complex*

$$G_{q,i} = L^{\beta_i}, \quad d_{G_q} = g d_{F_q} f,$$

where f and g act componentwise. Then:

- (1) G_q is a complex, $\text{rank}_A G_{q,i} = e(D)\beta_i$, and every differential entry has $\mathfrak{m}_A$ -adic order at least $q - C_0$ for a fixed C_0 ;
- (2) the maps

$$u_i = c^i f : G_{q,i} \longrightarrow F_{q,i}, \quad v_i = c^{d-i} g : F_{q,i} \longrightarrow G_{q,i}$$

are chain maps, and both degreewise composites are multiplication by $a = c^{d+1}$;

- (3) the cone of $u : G_q \rightarrow F_q$, regarded as a complex of A -modules, admits an explicit nullhomotopy for multiplication by a .

Proof. Write d_i for the differential of F_q . The composite $gd_i f$ first includes the chosen lattice into the corresponding free D -module, applies the Frobenius differential, and then multiplies by c to return to the lattice. It need not be the restriction of d_i itself, since d_i need not preserve L . Using $fg = cid_D$ and the A -linearity of all maps,

$$(gd_{F_q} f)^2 = gd_{F_q}(fg)d_{F_q} f = cgd_{F_q}^2 f = 0.$$

Minimality gives entries of d_F in $\mathfrak{n}$, so the entries of d_{F_q} belong to the Frobenius power $\mathfrak{n}^{[q]} \subseteq \mathfrak{n}^q$. If r is such an entry, its contribution to the compressed differential on b_j is

$$g(rb_j) = crb_j \in L \cap \mathfrak{n}^{q+a_j} D.$$

Formula (7.1) therefore gives order at least $q + a_j - a_\ell$ for its coefficient at b_ℓ . Thus one may take $C_0 = \max_{j,\ell} (a_\ell - a_j)$, which is independent of q and of the differential entry r . Since L is free of rank $e(D)$, L^{β_i} has rank $e(D)\beta_i$ as asserted.

The powers of c in the next maps compensate for the defect $fg = gf = c$. For example,

$$u_{i-1}d_{G_q} = c^{i-1}fgd_if = c^i d_if = d_i u_i,$$

and

$$d_{G_q} v_i = c^{d-i} g d_i f g = c^{d-i+1} g d_i = v_{i-1} d_i.$$

Thus both maps respect the differentials. Their composites are

$$v_i u_i = c^{d+1} \text{id}_{G_{q,i}}, \quad u_i v_i = c^{d+1} \text{id}_{F_{q,i}}.$$

With the cone convention

$$\text{Cone}(u)_i = F_{q,i} \oplus G_{q,i-1}, \quad d(y, x) = (d_F y + u_{i-1} x, -d_G x),$$

define $h_i(y, x) = (0, v_i(y))$. The chain-map identities give

$$dh(y, x) = (ay, -d_G v_i y), \quad hd(y, x) = (0, v_{i-1} d_F y + ax).$$

The two middle terms cancel because v is a chain map. Therefore $(dh + hd)(y, x) = (ay, ax)$, which is the required nullhomotopy with $a = c^{d+1}$. In particular, the scalar kills the identity map up to homotopy, a stronger assertion than its annihilating each homology module. $\square$

The compressed complex has fixed ranks $e(D)\beta_i$ over the regular ring A and differential depth $q - O(1)$. Its comparison with F_q is controlled by the fixed scalar c^{d+1} . To apply Theorem 6.3, it remains to replace the underlying A -complex F_q by uniformly bounded free models and to control its positive homology. We do this next.

8. DUTTA MULTIPLICITY OVER A SINGULAR COEFFICIENT RING

Let D be a complete local ring of characteristic $p > 0$ and dimension d . For a short free complex F with finite-length homology, its Dutta multiplicity is

$$\chi_\infty^D(F) = \lim_{n \rightarrow \infty} \frac{\chi(\mathcal{F}_D^n F)}{p^{nd}}, \quad \chi(X) = \sum_i (-1)^i \ell_D H_i(X). \quad (8.1)$$

The existence of the limit and its interpretation by local Riemann–Roch are due to Dutta and Roberts; see [6, 27] and [16, Sections 4.17–4.18]. The Frobenius functor here is extension of scalars: on a based free complex it raises the matrix entries to p^n -th powers. No residue-field factor occurs in this normalization.

The compressed complex of Proposition 7.2 is free over the regular parameter ring A , whereas its comparison target F_q need not be. The following replacement puts both complexes in the form required by Theorem 6.3.

Lemma 8.1 (bounded regular models). *In the situation of Proposition 7.2, the underlying A -complex F_q has a bounded free model Y_q whose total term rank and amplitude are bounded independently of q . The morphism $G_q \rightarrow F_q$ is represented by a chain map $G_q \rightarrow Y_q$ whose cone is nullhomotopic under c^{d+1} . Moreover,*

$$H_i(Y_q) \cong H_i(F_q), \quad \ell_A H_i(Y_q) = \ell_D H_i(F_q).$$

Proof. Since D is finite over the d -dimensional regular local ring A , choose once and for all a minimal A -free resolution $R_\bullet$ of D , in degrees $0, \dots, d$. We explain why the same resolution of the terms gives uniform control even though the differentials of F_q change with q .

Write $\beta_i = \text{rank}_D F_i$, and let $F_q^{\leq j}$ be the subcomplex with the same terms as F_q in degrees at most j and zero above degree j . Because the differentials lower degree, these are subcomplexes. They fit into short exact sequences of complexes

$$0 \longrightarrow F_q^{\leq j-1} \longrightarrow F_q^{\leq j} \longrightarrow D^{\beta_j}[j] \longrightarrow 0.$$

Start with the free model $R_\bullet^{\beta_0}$ of $F_q^{\leq 0}$. Suppose a free model of $F_q^{\leq j-1}$ has been constructed. Resolve the last term of the displayed sequence by $R_\bullet^{\beta_j}[j]$, and write U for the already constructed model of $F_q^{\leq j-1}$. The rotated connecting morphism has a chain-map representative

$$\alpha_j : R_\bullet^{\beta_j}[j-1] \longrightarrow U.$$

Its cone models the middle term $F_q^{\leq j}$. With our cone convention, the term in degree i is

$$\text{Cone}(\alpha_j)_i = U_i \oplus (R_\bullet^{\beta_j}[j-1])_{i-1} = U_i \oplus (R_\bullet^{\beta_j}[j])_i.$$

Thus the connecting map changes the differential but adds exactly the shifted resolution of the new term, with no further free summands.

After d steps this gives a free model of F_q in degrees $0, \dots, 2d$, with total term rank at most $(\sum_i \beta_i) \beta_A^{\text{tot}}(D)$. Cancelling unit entries produces a minimal model Y_q without increasing either its ranks or its amplitude. In particular,

$$\beta_A^{\text{tot}}(F_q) \leq \left(\sum_i \beta_i \right) \beta_A^{\text{tot}}(D).$$

Both bounds depend on the fixed A -module D and the ranks of F , and not on the powers appearing in the differential matrices.

We next transfer the comparison map and its nullhomotopy. Choose a quasi-isomorphism $Y_q \rightarrow F_q$. A bounded free source has the following lifting property: applying its Hom complex to a quasi-isomorphism gives a quasi-isomorphism of Hom complexes. To see this, apply Hom from each free term to the acyclic cone of the given quasi-isomorphism. Each such Hom complex is acyclic because Hom from a free module is exact. The filtration by source terms has these shifted complexes as its successive quotients; the long exact homology sequence, one quotient at a time, then shows that the whole Hom complex is acyclic. Thus chain maps lift up to homotopy, and a chain map whose image is nullhomotopic was already nullhomotopic.

Apply the lifting statement with source G_q . It provides a chain map $G_q \rightarrow Y_q$ whose composite to F_q is homotopic to the map in Proposition 7.2. The chosen homotopy gives a quasi-isomorphism from the new cone to the old cone. Apply the same Hom statement again, this time with source the new bounded free cone. The scalar endomorphism $c^{d+1} \text{id}$ maps to a nullhomotopic map on the old cone, by Proposition 7.2; it is therefore nullhomotopic on the new cone. This explains why the stronger scalar assertion survives the free replacement.

The quasi-isomorphism gives $H_i(Y_q) \cong H_i(F_q)$. Finally, A and D have the same residue field. A composition series of a finite-length D -module is also an A -composition series, which proves the equality of lengths. $\square$

We use the Frobenius homology estimate of Hochster–Huneke [13, Theorem 6.2], in the formulation restated by Bhatt–Hochster–Ma [3, Theorem 5.2]. Let X be a finite free complex in degrees $0, \dots, s$ with finite-length homology over a local ring of characteristic $p > 0$, and let M be a finite module of dimension d' . There is a constant C such that, for $0 \leq t \leq s$,

$$\ell H_{s-t}(M \otimes_D \mathcal{F}_D^n X) \leq Cp^{n \min\{d', t\}}. \quad (8.2)$$

The theorem requires no F -finiteness assumption. The specialization $M = D$ and $s = d$ used below is due to Roberts [26], as explicitly recorded by Bhatt–Hochster–Ma in the paragraph preceding their Theorem 5.2. Hochster–Huneke’s theorem provides the more general estimate stated here. Taking $M = D$, $X = F$, $s = d$, and $t = d - i$ gives

$$\ell_D H_i(F_q) = O(q^{d-i}), \quad q = p^n. \quad (8.3)$$

For every positive i , this exponent is at most $d - 1$. Summing over the finitely many positive degrees therefore gives an error of order $O(q^{d-1})$, which disappears after normalization by q^d . The degree-zero homology is the only term that can contribute on that scale.

Proof of Theorem 1.2. First assume that $d > 0$ and that the residue field is infinite. Apply Propositions 7.1 and 7.2 to F_q , and take the free models from Lemma 8.1. We check the hypotheses of Theorem 6.3 over the regular ring A . The free complexes G_q are supported in degrees $0, \dots, d$ and have fixed ranks $b_i = e(D)\beta_i$, with $b_0 > 0$ by the hypothesis $F_0 \neq 0$. For sufficiently large q their differentials have depth at least $N_q = q - C_0 > 0$, so $N_q/q \rightarrow 1$. The models Y_q have uniformly bounded ranks and amplitude, and their homology agrees with that of F_q ; it is finite-length and vanishes in negative degrees. The comparison cone is nullhomotopic under the fixed nonzero scalar c^{d+1} . Its order is independent of q , and in particular is $o(q)$. Finally, equation (8.3), together with the equality of lengths over A and D , gives

$$\sum_{j>0} \ell_A H_j(Y_q) = O(q^{d-1}) = o(q^d).$$

The scalar-class mass theorem, with $\rho = 1$, now yields

$$\liminf_q \frac{\ell_D H_0(F_q)}{q^d} \geq \frac{e(D)\beta_i}{\binom{d}{i}}. \quad (8.4)$$

More explicitly, the same homology estimate gives

$$|\chi(F_q) - \ell_D H_0(F_q)| \leq \sum_{j>0} \ell_D H_j(F_q) = O(q^{d-1}).$$

The normalized Euler characteristics have a limit by the definition and existence theorem for Dutta multiplicity. Since the displayed difference divided by q^d tends to zero, the normalized zeroth homology lengths have the same limit. Hence

$$\chi_\infty^D(F) = \lim_q \frac{\chi(F_q)}{q^d} = \lim_q \frac{\ell_D H_0(F_q)}{q^d}.$$

Together with (8.4), this proves the desired inequality.

For a finite residue field k , introduce an indeterminate z and set

$$D' = \widehat{D[z]_{\mathfrak{m}D[z]}}.$$

The map $D \rightarrow D'$ is flat and local, its maximal ideal is $\mathfrak{m}D'$, and its residue field is the infinite field $k(z)$. This extension preserves the lengths of finite-length modules: tensor a composition series with D' , and each factor k becomes one copy of the residue field $k(z)$. In particular, it preserves every value of the Hilbert–Samuel function $\ell(D/\mathfrak{m}^r)$ and hence both dimension and multiplicity. A minimal free complex stays minimal because its differential entries remain in the extended maximal ideal; its free ranks are unchanged. Finally, applying the ring map to a matrix entry and then raising it to its p^n -th power gives the same result as doing these operations in the opposite order. Thus Frobenius commutes with this extension on based free complexes:

$$\mathcal{F}_{D'}^n(F \otimes_D D') \cong \mathcal{F}_D^n(F) \otimes_D D'.$$

Flat base change therefore preserves the homology lengths of every Frobenius pullback, and thus preserves χ_∞ . Applying the infinite-residue-field case over D' proves the result over D .

If $d = 0$, the short complex is concentrated in degree zero. Each Frobenius pullback is the free module D^{β_0} , so $\chi_\infty^D(F) = \ell(D)\beta_0 = e(D)\beta_0$. $\square$

Corollary 8.2 (passage to completion). *Let $(D, \mathfrak{m}, k)$ be a Noetherian local ring of characteristic $p > 0$ and dimension d , and let F be a minimal complex of finite free D -modules in degrees $0, \dots, d$. Assume that its homology has finite length and that $F_0 \neq 0$. The limit in (8.1) exists over D , and for every $0 \leq i \leq d$,*

$$\binom{d}{i} \chi_\infty^D(F) \geq e(D) \operatorname{rank}_D F_i.$$

Thus completeness is unnecessary for the inequality.

Proof. Put $\widehat{F} = F \otimes_D \widehat{D}$, where $\widehat{D}$ is the $\mathfrak{m}$ -adic completion. The map $D \rightarrow \widehat{D}$ is faithfully flat, its maximal ideal is $\mathfrak{m}\widehat{D}$, and its residue field is k . Tensoring a composition series therefore shows that

$$\ell_{\widehat{D}}(L \otimes_D \widehat{D}) = \ell_D L$$

for every finite-length D -module L . Applying this to $D/\mathfrak{m}^r$ shows that completion preserves the Hilbert–Samuel function, hence dimension and multiplicity. Flatness identifies the homology of $\widehat{F}$ with the completion of the homology of F . Moreover, $\widehat{F}$ is minimal, has the same free ranks as F , and has $\widehat{F}_0 \neq 0$. It consequently satisfies the hypotheses of Theorem 1.2.

We also check the invariant on the left side. In fixed free bases, Frobenius pullback raises each differential entry to its p^n -th power. The completion homomorphism commutes with these powers, so

$$\mathcal{F}_{\widehat{D}}^n(\widehat{F}) \cong (\mathcal{F}_D^n F) \otimes_D \widehat{D}.$$

The homology of $\mathcal{F}_D^n F$ has finite length: away from $\mathfrak{m}$, the complex F is a bounded exact complex of free modules, hence split exact, and Frobenius preserves its splitting. Flatness and the equality of finite lengths now give, for every n ,

$$\chi^{\widehat{D}}(\mathcal{F}_{\widehat{D}}^n \widehat{F}) = \chi^D(\mathcal{F}_D^n F).$$

The normalized sequence on the left converges by the complete-ring existence theorem used in (8.1). Thus the sequence on the right converges as well, and $\chi_\infty^{\widehat{D}}(\widehat{F}) = \chi_\infty^D(F)$. Theorem 1.2 over $\widehat{D}$, together with the invariance of multiplicity and ranks, proves the assertion. $\square$

9. FLAT LOCAL MAPS

The remaining steps use reductions from the literature. We separate them from the preceding estimates to make clear which part of the argument is supplied by earlier work. The role of the first reduction is to encode the target multiplicity as the Dutta multiplicity of a short complex over a ring having the source multiplicity. The degree-zero case of Theorem 1.2 can then compare the two numbers directly.

Proposition 9.1 (Ma; Iyengar–Ma–Walker). *Let $(R, \mathfrak{m}) \rightarrow (S, \mathfrak{n})$ be a flat local map of characteristic $p > 0$. To prove $e(R) \leq e(S)$ for all such maps, it suffices to treat the case in which R and S are complete of the same dimension, R is a domain, and the residue field of S is algebraically closed. In that case there are a complete local ring $(T, \mathfrak{q})$ and a minimal short T -free complex X such that*

$$e(T) = e(R), \quad X_0 = T, \quad \chi_\infty^T(X) = e(S). \quad (9.1)$$

The residue field of T is the residue field of S .

Proof. The first assertion is [21, Lemma 2.2]. Its proof uses completion, localization in the target, prime filtrations, and a flat extension of the target residue field. The reduction preserves the implication to be proved and stays in characteristic p ; it does not require S to be a domain. In particular, this is a reduction of the universal inequality: proving it for the stated restricted class of maps proves it for every original map. We do not assert that every individual operation preserves each multiplicity separately.

For the second assertion we use the construction in the proof of [16, Proposition 8.3]. A complete Cohen factorization [1, Theorem 1.1] gives

$$R \longrightarrow T \longrightarrow S = T/J,$$

where $R \rightarrow T$ is flat with regular closed fiber. Thus $e(T) = e(R)$, and the factorization also gives $\mathrm{pd}_T S < \infty$ because the original map is flat. Lemma 3.5 of [21], applied under these hypotheses, shows that J is perfect. Put $s = \dim S$ and $h = \mathrm{ht} J$; then $\dim T = s + h$. Choose $\mathbf{x} = x_1, \dots, x_s$ in $\mathfrak{q}$ whose images generate a minimal reduction of $\mathfrak{n}$ and which form part of a system of parameters of T . If P is the minimal T -free resolution of S , then P has length h , and

$$X = K(\mathbf{x}; T) \otimes_T P$$

is a free complex in degrees $0, \dots, s + h = \dim T$. Both factors have differential entries in $\mathfrak{q}$, so their tensor product is minimal. Also $P_0 = T$, since $S = T/J$ is cyclic, and hence $X_0 = T$.

The augmentation $P \rightarrow S$ induces a quasi-isomorphism $X \rightarrow K(\mathbf{x}; S)$: tensoring with the bounded free Koszul complex preserves the quasi-isomorphism. The images of $\mathbf{x}$ generate an $\mathfrak{n}$ -primary ideal, so the Koszul homology is supported only at $\mathfrak{n}$ and has finite length. The parameter Koszul formula identifies its Euler characteristic with the multiplicity of that parameter ideal. Since this ideal is a reduction of $\mathfrak{n}$, this is $e(S)$. The residue fields of T and S agree, so the homology lengths give the same Euler characteristic over either ring. We have therefore obtained $\chi^T(X) = e(S)$.

The equality $\chi_\infty^T(X) = \chi^T(X)$ is the final claim proved in [16, Proposition 8.3]. Its proof localizes at the primes of T lying over the top-dimensional minimal primes of S . These local rings are complete intersections, by Tabaa's theorem on the fibers of $R \rightarrow T$ [31, Théorème 2]; compatibility of local Riemann–Roch with localization then identifies the two Euler characteristics for this particular complex. This identification is an additional ingredient of their construction: the Koszul Euler characteristic computation alone does not identify Dutta multiplicity over a singular ring. Applying that result here gives $\chi_\infty^T(X) = e(S)$ and proves (9.1) without imposing a Cohen–Macaulay hypothesis on T . $\square$

Theorem 9.2. *Lech's inequality holds for every flat local map of Noetherian local rings of characteristic $p > 0$.*

Proof. Apply Proposition 9.1 and take the ring T and complex X supplied there. The degree-zero inequality in Theorem 1.2 gives

$$e(S) = \chi_\infty^T(X) \geq e(T) \mathrm{rank}_T X_0 = e(T) = e(R). \quad \square$$

Proposition 9.3 (Reduction to positive characteristic). *Suppose Lech's inequality holds for all flat local maps in positive characteristic and in every dimension. Then it holds for all flat local maps of equal characteristic zero.*

Proof. This is the consequence with constant $C = 1$ of the reduction argument in [21, Section 5]. Ma attributes the strategy to Hochster. Although that section applies the argument to a different numerical bound, its construction preserves the two multiplicities separately, so it applies unchanged to Lech's inequality. We recall the points that ensure this.

By [21, Lemma 5.1], a characteristic-zero counterexample can be replaced by a finite free local map between complete rings with a common algebraically closed coefficient field. Section 5.1

of [21] uses Artin approximation [2] to descend this finite diagram to local rings essentially of finite type. The equations record the algebra structures, the finite free map, minimal reductions of both maximal ideals, and the homology of their parameter Koszul complexes. In particular, each multiplicity is preserved. The numerical reason for recording this homology is the parameter Koszul formula: for a minimal reduction $\mathbf{r}$ in a local ring R , it expresses

$$e(R) = \sum_i (-1)^i \ell_R H_i(K(\mathbf{r}; R)).$$

There is a corresponding formula for the chosen minimal reduction in S . Preserving these two finite lists of homology lengths preserves $e(R)$ and $e(S)$ individually.

Section 5.2 of [21] spreads the diagram over a finitely generated $\mathbb{Z}$ -algebra and applies generic freeness so that the relevant Koszul homology commutes with specialization and retains its length. The resulting positive-characteristic flat local map has the same two multiplicities. A strict inequality $e(R) > e(S)$ would therefore survive specialization, contradicting the hypothesis. $\square$

Proof of Theorem 1.1. In positive characteristic use Theorem 9.2. In characteristic zero apply Proposition 9.3 to that theorem. $\square$

10. SCOPE OF THE ARGUMENT

Equal characteristic is used in the parameter lattice (Proposition 7.1), in the opposite x - and u -gradings of the degree boxes (Proposition 5.1), and in the Frobenius family controlled by (8.3). Characteristic zero follows by specialization. The argument provides no analogous family in mixed characteristic: the integer p has degree zero in any unital ring grading, and reduction modulo p need not preserve the required homology or either multiplicity.

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